Spectrum and frame: a workbook

Part I: One PSD matrix, power-law spectra, and what they say about eigenvectors

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Version with solutions (Appendix B)

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How to use this workbook

This workbook is the learning version of `spectral_frame.pdf` (Part I). It covers the same mathematics, adds the background each part needs, and turns the results into exercises that you prove yourself. The compact note stays as a reference: read it afterwards to see the whole argument in fourteen pages.

Kinds of items.

Fact: a background result you may use without proof. Each comes with a reading pointer.

Warm-up: a few lines. It checks that a definition has landed. Do all of them.

Exercise: a standard proof or computation, usually less than half a page.

Guided proof: a result broken into steps (a), (b), ... Each step is a small exercise. Later steps may use earlier ones, even if you skipped them, so you can enter a proof in the middle.

Compute: a short numpy experiment that runs in under a minute on a laptop. The expected numbers are given; `make_figures.py` in this folder is a reference solution.

Reading and **Lab connection** boxes say where to read the background and what the lab measured. Lab results name the study; [L] marks local measurements at fixed training states.

Hints for every item are in Appendix A, in order. Full solutions are in a separate file, the same document with an Appendix B: `workbook_solutions.pdf`. Try a step for a while before you look.

Prerequisites. Eigenvalues and eigenvectors, orthogonal matrices and the trace; Gaussian random vectors, expectation and variance; integrals and first-order Taylor expansion. Everything else is introduced where it is needed.

Route. Sections 1 and 2 are the base. Sections 3, 4 and 5 are independent of each other and can be read in any order. Section 6 uses all of them and opens Part II.

Why this matters. Three results from the lab's CIFAR-10 studies (a depth-6 ViT; per-tensor generalized Gauss–Newton (GGN) blocks) motivate the notes.

(1) Late in training the per-tensor GGN spectrum is a nearly fixed power law with exponent $\alpha \approx 1$, while its eigenvectors rotate. A top-16 subspace cached 10 steps earlier keeps only 0.84–0.88 of the current top-16 curvature (`spectral_rotation_20261020`, [L]).

(2) A spectrum alone cannot precondition the bulk: a power-law diagonal reaches only the level of the exact diagonal. A few curvature products can (`bulk_krylov_20261014`, [L]).

(3) The best power of the curvature falls from about 3/4 early in training to 1/2 late, as the noise scales of strongly preconditioned steps grow (`noise_scale_20261021`, [L]).

SOAP already has the structure that the first result suggests: an eigenbasis refreshed rarely and a diagonal second-moment estimate refreshed every step. The question behind these notes is whether a model of spectrum and frame can explain that design and improve it. Part I treats one fixed matrix. By the end you will have proved three things. The spectrum is a probability distribution, and it fixes every rotation-invariant quantity. It says nothing about the frame. And a power law implies that eigenvalues survive perturbations much deeper down the spectrum than eigenvectors do. That asymmetry is the mathematical reason why "stable spectrum, rotating frame" is the natural picture.

Notation. $\mathbb{S}^D$ is the set of real symmetric $D \times D$ matrices and $\mathbb{S}^D_+$ its positive semi-definite (PSD) subset; $\mathrm{O}(D)$ is the group of orthogonal matrices. Eigenvalues are sorted, $\lambda_1 \geq \cdots \geq \lambda_D$. $\|\cdot\|_{\mathrm{op}}$ is the operator norm (largest singular value) and $\|\cdot\|_{\mathrm{F}}$ the Frobenius norm. δ_x is the point mass at x. $H_n = \sum_{k \leq n} 1/k$ is the harmonic number; we use $H_n = \ln n + \gamma + O(1/n)$ with Euler's constant $\gamma \approx 0.5772$. $\zeta(s) = \sum_{k \geq 1} k^{-s}$ is the Riemann zeta function, and $\zeta(2) = \pi^2/6$.

1 Spectrum and frame

A symmetric matrix is a rotated diagonal matrix. This section separates the two parts of that sentence: the diagonal, which we call the *spectrum*, and the rotation, which we call the *frame*. We count how many numbers each part holds, and we ask which quantities depend on which part. The answers are short, but every later section uses them.

> **Reading.** Strang (2019), Part I, explains the spectral theorem and functions of matrices in the language of data. Horn & Johnson (2013), Sections 2.5 (the spectral theorem) and 4.1 (Hermitian matrices), are the standard reference. Golub & Van Loan (2013), Section 8.1, is a compact summary that also contains the perturbation results of Section 5.

1.1 The spectral theorem and functions of a matrix

Fact 1.1 (Spectral theorem)**.** *Every $A \in \mathbb{S}^D$ can be written $A = U \Lambda U^\top$ with $U \in \mathrm{O}(D)$ and $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_D)$ real. The columns u_k of U are orthonormal eigenvectors, $Au_k = \lambda_k u_k$, and $A = \sum_k \lambda_k u_k u_k^\top$.*

Definition 1.2 (Spectrum, frame, functions of a matrix)**.** Sort the eigenvalues, $\lambda_1 \geq \cdots \geq \lambda_D$. The vector $\sigma(A) = (\lambda_1, \ldots, \lambda_D)$ is the *spectrum* of A, and any U as in Fact 1.1 is a *frame* of A. A is *positive semi-definite* (PSD) if $v^\top A v \geq 0$ for all v. For a function f defined on the spectrum, $f(A) = U \,\mathrm{diag}(f(\lambda_1), \ldots, f(\lambda_D))\, U^\top$.

The definition of $f(A)$ hides a question. If eigenvalues repeat, the frame is not unique, so is $f(A)$ well defined? Exercise 1.5 answers it.

Warm-up 1.3 (What the spectrum alone determines)**.** Let $A = U \Lambda U^\top$. Show:

(a) $\mathrm{tr}\, A = \sum_k \lambda_k$ and $\|A\|_{\mathrm{F}}^2 = \sum_k \lambda_k^2$;

(b) $\|A\|_{\mathrm{op}} = \max_k |\lambda_k|$;

(c) $\lambda_D \|v\|^2 \leq v^\top A v \leq \lambda_1 \|v\|^2$ for all v, with equality at $v = u_D$ and at $v = u_1$;

(d) A is PSD if and only if $\lambda_D \geq 0$.

Warm-up 1.4. If $Au = \lambda u$ and $Aw = \mu w$ with $\lambda \neq \mu$, then $u^\top w = 0$.

Exercise 1.5 (A function of a matrix does not depend on the frame)**.** (a) Let p be a polynomial with $p(\lambda_k) = f(\lambda_k)$ for every k. It exists by Lagrange interpolation on the distinct values among the λ_k. Show $U \,\mathrm{diag}(f(\lambda_1), \ldots, f(\lambda_D)) U^\top = p(A)$.

(b) Conclude that $f(A)$ is the same for every frame of A.

(c) Show that $f(A)$ commutes with A. Check that A^2, $(A + \delta I)^{-1}$ (for $\delta > -\lambda_D$) and, for PSD A, $A^{1/2}$ and $(A + \delta I)^{-\nu}$ (for $\delta > 0$) are functions of A in this sense.

Warm-up 1.6 (Rotating a matrix)**.** Let $Q \in \mathrm{O}(D)$. Show that $\|Qv\| = \|v\|$, that $QAQ^\top$ has the spectrum of A and the eigenvectors Qu_k, and that $Qf(A)Q^\top = f(QAQ^\top)$.

1.2 How many numbers are in the spectrum, and how many in the frame

A symmetric matrix has $D(D+1)/2$ free entries. The spectrum accounts for D of them, so the frame should account for the remaining $D(D-1)/2$. The guided proof below makes this count precise. It finds all directions in which a matrix can move while its spectrum stays fixed.

Definition 1.7 (Orbit of a spectrum)**.** For a fixed diagonal Λ, the *orbit* $\mathcal{O}_\Lambda = \{Q\Lambda Q^\top : Q \in \mathrm{O}(D)\}$ is the set of symmetric matrices with the spectrum of Λ (by Fact 1.1 and Warm-up 1.6).

Guided proof 1.8 (The orbit has $D(D-1)/2$ directions)**.** Let $\Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_D)$.

(a) Let $Q(t)$ be a differentiable curve in $\mathrm{O}(D)$ with $Q(0) = I$. Differentiate $Q(t)^\top Q(t) = I$ and show that $W = Q'(0)$ is skew-symmetric, $W^\top = -W$. Conversely, show that $Q(t) = e^{tW}$ is orthogonal for every skew-symmetric W.

(b) The skew-symmetric $D \times D$ matrices form a vector space of dimension $D(D-1)/2$.

(c) Along $A(t) = Q(t)\Lambda Q(t)^\top$, show $A'(0) = W\Lambda - \Lambda W$.

(d) Show $(W\Lambda - \Lambda W)_{ij} = (\lambda_j - \lambda_i)W_{ij}$. In particular the diagonal of $A'(0)$ is zero.

(e) Suppose the λ_k are distinct. Show that $W \mapsto W\Lambda - \Lambda W$ is injective on skew-symmetric matrices, and that its image is the space of symmetric matrices with zero diagonal. Conclude that Λ can move along its orbit in $D(D-1)/2$ independent directions, and that these together with the D diagonal directions (changes of the spectrum) span all of $\mathbb{S}^D$.

(f) If the distinct values have multiplicities $m_1, \dots, m_r$ (and Λ lists equal values next to each other), show that the kernel of $W \mapsto W\Lambda - \Lambda W$ consists of the block-diagonal skew-symmetric matrices, of dimension $\sum_j m_j(m_j-1)/2$. The orbit then has $D(D-1)/2 - \sum_j m_j(m_j-1)/2$ directions.

(g) (Uniqueness of the frame.) If the λ_k are distinct and $Q\Lambda Q^\top = \Lambda$ for some $Q \in \mathrm{O}(D)$, show that Q is diagonal with entries ± 1. Hence the frame of a matrix with distinct eigenvalues is unique up to the signs of its columns.

Remark 1.9 (The seed of Part II). Step 1.8(d) is the most important formula in this workbook. Written in the eigenbasis, a rotation of the frame changes only off-diagonal entries, and a change of the spectrum changes only diagonal entries. To produce an off-diagonal entry of size x at position (i, j), the frame must rotate by $x/(\lambda_j - \lambda_i)$: small gaps mean large rotations. The same formula returns as first-order perturbation theory (Guided proof 5.13) and as the equation of motion of the frame (Guided proof 6.1).

> **Lab connection** ([L]). For the 64×64 weight blocks of the lab's ViT ($D = 4{,}096$) the spectrum is 4,096 numbers and the frame 8,386,560. For the 64×256 blocks ($D = 16{,}384$) they are 16,384 and 134,209,536. A power-law fit compresses the spectrum to two or three numbers. "The spectrum is stable" therefore constrains a vanishing part of the matrix; almost all of its information is in the frame.

1.3 Invariant and equivariant maps

Some quantities do not depend on the frame at all, and some maps rotate along with the matrix. These two classes organise the rest of the workbook. Everything a power law says belongs to the first class, and every preconditioner that needs no stored eigenvectors belongs to the second.

Definition 1.10 (Invariant and equivariant)**.** $F : \mathbb{S}^D \to \mathbb{R}$ is *invariant* if $F(QAQ^\top) = F(A)$ for all $Q \in \mathrm{O}(D)$. $G : \mathbb{S}^D \to \mathbb{S}^D$ is *equivariant* if $G(QAQ^\top) = Q\,G(A)\,Q^\top$ for all $Q \in \mathrm{O}(D)$.

Exercise 1.11. Show that F is invariant if and only if $F(A)$ depends on A only through $\sigma(A)$.

Warm-up 1.12 (Examples)**.** (a) Which of these are invariant: $\operatorname{tr} A$, $\det A$, $\lambda_1(A)$, A_{11}, $\|A\|_{\mathrm{F}}$, and $v^\top A v$ for a fixed v?

(b) Which of these are equivariant: A^2; $(A+\delta I)^{-1}$; the diagonal part $\operatorname{Diag}(A)$, which keeps the entries $A_{11},\dots,A_{DD}$ and sets all others to zero; the projector $\Pi_k(A)$ onto the top-k eigenvectors, on matrices with $\lambda_k > \lambda_{k+1}$; and $A + B_0$ for a fixed matrix B_0?

Guided proof 1.13 (Equivariant maps are functions of the matrix)**.** Let G be equivariant and $A = \sum_k \lambda_k u_k u_k^\top$.

(a) Show that the reflection $R_k = I - 2u_k u_k^\top$ is orthogonal and symmetric, that $R_k u_k = -u_k$ and $R_k u_j = u_j$ for $j \neq k$, and that the eigenspace of R_k for the eigenvalue -1 is $\operatorname{span}(u_k)$.

(b) Show $R_k A R_k^\top = A$.

(c) Deduce $R_k G(A) R_k = G(A)$.

(d) Apply both sides of (c) to u_k and conclude that $G(A)u_k = \varphi_k u_k$ for some number φ_k.

(e) Conclude $G(A) = \sum_k \varphi_k u_k u_k^\top$, so $G(A)$ commutes with A. If the eigenvalues are distinct, show $G(A) = p(A)$ for a polynomial p (which may depend on A).

Exercise 1.14 (Krylov methods are jointly equivariant)**.** Let $F(A,b) = p_{A,b}(A)\,b$, where $p_{A,b}$ is a polynomial whose coefficients depend on (A,b) only through the numbers $b^\top A^i b$, $i = 0,1,2,\dots$. Show $F(QAQ^\top, Qb) = Q\,F(A,b)$ for all $Q \in \mathrm{O}(D)$. Conjugate gradients and Lanczos produce iterates of this form (Section 4.3). By contrast, consider $x = U_{\mathrm{old}} \operatorname{diag}(\varphi) U_{\mathrm{old}}^\top b$ with a stored frame U_{old}. What happens to x if you rotate A and b but not U_{old}?

Remark 1.15 (What this means for us). Every statement about "the spectrum" (a power law, a trace, a condition number) is an invariant function: it takes the same value on the whole orbit and says nothing about the frame. A preconditioner that must not depend on an unknown frame must be equivariant, hence a function of the current matrix, and it can be applied with matrix–vector products alone. Krylov methods are of this kind. A stored eigenbasis (SOAP, PARSEC-L) is the other choice: it carries frame information from the past into the present, and Section 5 asks when that information is still valid.

2 Power laws: four faces

The lab fits straight lines to log–log plots of sorted eigenvalues against their rank. This section shows that such a plot describes a probability distribution, read from its top, and that a power law can be stated in four equivalent ways. Each answers a different question: how large is the k-th eigenvalue (*rank*); how many eigenvalues exceed a threshold (*counting*); which eigenvalue does a randomly chosen eigen-direction have (*distribution*); and how many directions lie within a given depth below the top (*depth*).

> **Reading.** Newman (2005), Sections 1–2, compares rank plots with histograms and explains why cumulative plots are preferred; that is exactly the rank/counting equivalence below. Clauset, Shalizi & Newman (2009) show how straight lines on log–log plots mislead, and how to fit and test power laws properly. Martin & Mahoney (2021) study heavy-tailed spectra of trained weight matrices, and Xie et al. (2022) power-law Hessian spectra; both are ML readings for this section.

2.1 Spectral distribution, counting function and rank function

Definition 2.1. The *empirical spectral distribution* (ESD) of A is the probability measure $\mu_A = \frac{1}{D}\sum_{k=1}^{D} \delta_{\lambda_k}$. Its *counting function* is $N_A(t) = \#\{k : \lambda_k \geq t\} = D\,\mu_A([t,\infty))$. The *rank function* is $k \mapsto \lambda_k$.

Think of μ_A as the law of the eigenvalue of a uniformly chosen eigen-direction. Section 4 shows that it is also the law of the curvature along a random direction, which is the reason to care about it.

Warm-up 2.2. Let $A = \operatorname{diag}(3, 1, 1, 0)$. Write down μ_A, sketch $N_A(t)$ for all real t, and list the rank function. Check that $N_A(t) \geq k$ holds exactly when $\lambda_k \geq t$, for $k = 2, 3, 4$ and $t = 1$.

Exercise 2.3 (A rank plot is a quantile plot)**.** (a) Show that $\lambda_k \geq t$ if and only if $N_A(t) \geq k$.

(b) Deduce $\lambda_k = \max\{t : N_A(t) \geq k\}$.

(c) Define the upper quantile function $Q_A(p) = \max\{t : \mu_A([t, \infty)) \geq p\}$ for $0 < p \leq 1$. Show $\lambda_k = Q_A(k/D)$.

So a log–log rank plot is a log–log plot of the upper quantiles of μ_A.

2.2 A power law and its four faces

Definition 2.4 (Power law on a rank window)**.** *A* follows a *power law* with exponent $\alpha > 0$ and scale $c > 0$ on the window $\{1, \dots, K\}$, $K \leq D$, if $\lambda_k = c\,k^{-\alpha}$ for $k \leq K$. The law is *pure* if $K = D$. In data the statement is statistical: $\log \lambda_k = \log c - \alpha \log k + r_k$ with small residuals r_k, fitted on a chosen window.

Fact 2.5 (Pareto distribution)**.** *A random variable X is Pareto with* tail index $a > 0$ *and minimum $x_0 > 0$ if $\Pr(X \geq t) = (t/x_0)^{-a}$ for $t \geq x_0$. Its density is $a\,x_0^a t^{-a-1}$ on $[x_0, \infty)$, and $\mathbb{E}X^q < \infty$ if and only if $q < a$.*

Guided proof 2.6 (Four faces of a power law)**.** Let $\lambda_k = c\,k^{-\alpha}$ for $k \leq K$, with $\lambda_{K+1} < \lambda_K$ if $K < D$. Fix t with $\lambda_K \leq t \leq c$.

(a) Show that no rank $k > K$ is counted in $N_A(t)$.

(b) For $k \leq K$, show that $\lambda_k \geq t$ if and only if $k \leq (c/t)^{1/\alpha}$, and that $(c/t)^{1/\alpha} \leq K$.

(c) *Counting face.* Conclude $N_A(t) = \lfloor (c/t)^{1/\alpha} \rfloor$.

(d) *Distribution face.* Deduce $\mu_A([t, \infty)) = \lfloor (c/t)^{1/\alpha} \rfloor / D$. Ignoring the floor, compare with Fact 2.5: on $[\lambda_K, c]$, μ_A looks like the upper tail of a Pareto law with tail index $1/\alpha$, carrying mass K/D. Show that the smoothed density is $\rho(t) = c^{1/\alpha} t^{-1-1/\alpha}/(\alpha D)$.

(e) *Depth face.* Let $v = \ln(c/t)$ be the depth below the top eigenvalue, measured in nats. Show that the number of eigenvalues within depth v is $\lfloor e^{v/\alpha} \rfloor$ for $0 \leq v \leq \alpha \ln K$. A power law in rank is an exponential law in log-depth.

(f) *Converse.* Suppose $N_A(t) = \lfloor (c/t)^{1/\alpha} \rfloor$ for all $t \geq c\,K^{-\alpha}$. Use Exercise 2.32.3(b) to show $\lambda_k = c\,k^{-\alpha}$ for $k \leq K$.

Exercise 2.7 (Reading the exponent)**.** (a) Show $\lambda_{2k}/\lambda_k = 2^{-\alpha}$: doubling the rank lowers the eigenvalue by α octaves.

(b) Show that going one octave deeper ($t \mapsto t/2$) multiplies $N_A(t)$ by $2^{1/\alpha}$, up to rounding. For $\alpha = 1$ each octave holds twice as many directions as the one above it, so most directions sit near the bottom (Figure 1c).

(c) Take a pure law with $\alpha = 1$ and $D = 4{,}096$. What percentage of the eigen-directions lies within a factor 10 of λ_1, and what percentage within a factor 1,000? Which share of the trace does each group carry? (Answers: 0.24% of the directions carry 0.329 of the trace; 24% carry 0.842.)

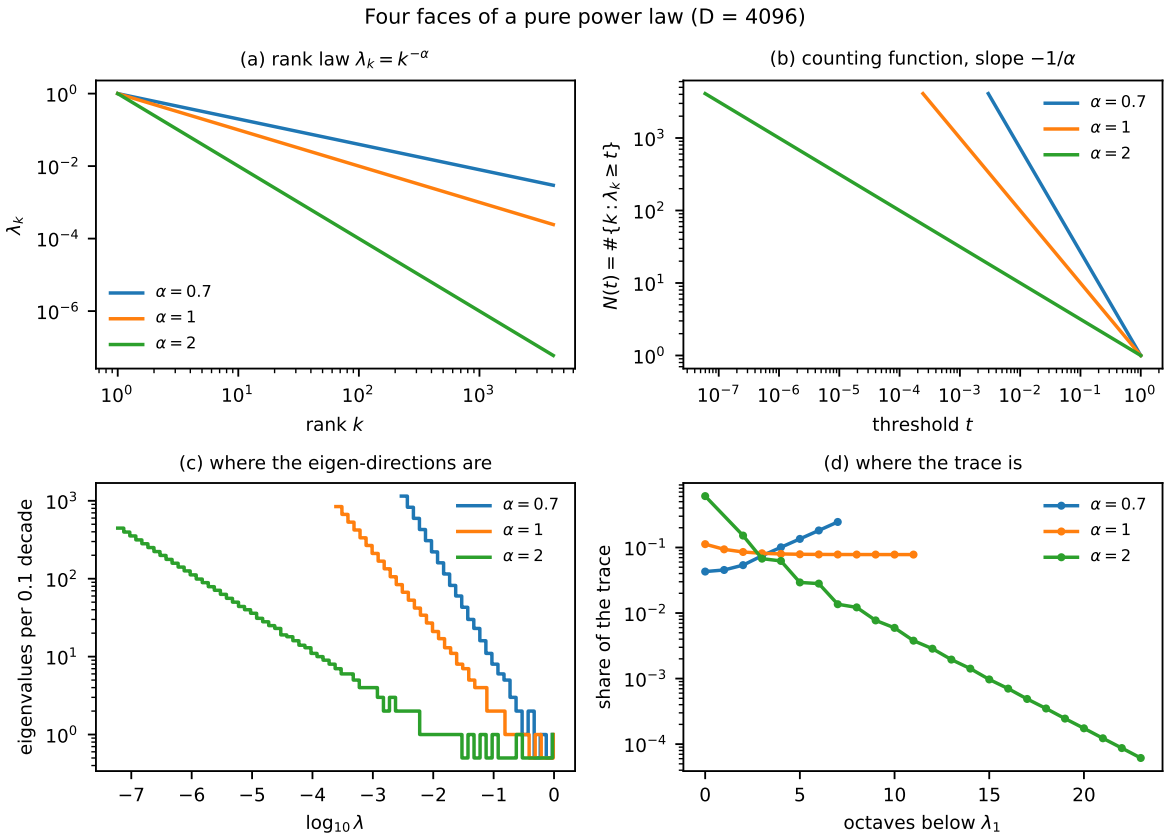


Figure 1: Four faces of a pure power law $\lambda_k = k^{-\alpha}$, $D = 4{,}096$ (synthetic). (a) The rank law. (b) Its counting function, with slope $-1/\alpha$ (Guided proof 2.6). (c) Eigenvalues per 0.1 decade: most eigen-directions sit near the bottom (Exercise 2.7). (d) Share of the trace per full octave below λ_1: top-heavy for $\alpha = 2$, flat for $\alpha = 1$, bottom-heavy for $\alpha = 0.7$ (Exercise 3.4).

Exercise 2.8 (Transformations of a law)**.** (a) Show that $A \mapsto sA$ $(s > 0)$ changes c to sc and keeps α: the exponent is scale-free.

(b) Show that $A \mapsto A^p$ $(p > 0$, A PSD) gives exponent αp and scale c^p.

(c) Precondition $A \succ 0$ by a power of itself, $P = A^{-\nu}$ with $0 \leq \nu < 1$. Show $P^{1/2} A P^{1/2} = A^{1-\nu}$, and give its exponent and its condition number λ_1/λ_K on the window. (This is the fractional-Newton family of `noise_scale_20261021`: $\nu = 0$ is the gradient step and $\nu = 1$ Newton.)

(d) Now precondition with a function of a *different* matrix. Let $A = \mathrm{diag}(1, \varepsilon)$ with $0 < \varepsilon < 1$, and $P = B^{-1}$ with $B = \mathrm{diag}(\varepsilon, 1)$: the same spectrum in the swapped frame. Compute the condition number of $P^{1/2} A P^{1/2}$ and compare it with that of A. What does this say about statements like (c) when P has a different frame from A?

Remark 2.9 (Head law and global law). A fit on ranks $1, \ldots, K$ with $K \ll D$ describes only the top K/D of μ_A. The lab fits ranks 1–200 in blocks with D between 640 and 16,384, that is, the top 31% down to the top 1.2% of the eigen-directions. The rest of μ_A, where most directions live, is not covered by the fit. Section 3 works out which quantities the fit nevertheless decides.

Exercise 2.10 (Broken laws)**.** Let $\lambda_k = c\,k^{-\alpha_1}$ for $k \leq k_b$ and $\lambda_k = c\,k_b^{-\alpha_1}(k/k_b)^{-\alpha_2}$ for $k_b < k \leq K$, so the law is continuous at the break k_b. Write down $N_A(t)$ on $[\lambda_K, c]$, ignoring rounding. Show that on this range μ_A consists of two Pareto pieces, and give their tail indices.

Lab connection ([L]). Early in training the lab's whitened per-tensor GGN spectra are broken. Ranks 1–16 fall roughly as k^{-1} and ranks 17–200 with $\alpha_{\text{bulk}} = 1.79$–$2.10$ at steps 10–25; the break flattens by step 100. A single law extrapolated from the top 16 then underestimates λ_{17} by a factor 1.6–2.4 (`bulk_krylov_20261014`). The raw per-tensor GGN fits on ranks 1–200 go from $\alpha \approx 1.8$–2.1 at step 10 to $\alpha \approx 0.9$–1.1 at steps 250–500, with $R^2 \approx 0.99$ (`spectral_rotation_20261020`).

2.3 Kronecker products bend a law

SOAP's eigenbasis is the Kronecker product of the eigenbases of two small factor matrices. This subsection shows that exact power laws in the two factors do not give an exact power law in the product.

Warm-up 2.11 (Eigenpairs of a Kronecker product)**.** For $R \in \mathbb{S}^n$ and $L \in \mathbb{S}^m$, show $(R \otimes L)(v \otimes w) = (Rv) \otimes (Lw)$. Deduce: if $Rv_j = r_j v_j$ and $Lw_i = \ell_i w_i$, then $v_j \otimes w_i$ is an eigenvector of $R \otimes L$ with eigenvalue $r_j \ell_i$, and these mn vectors are orthonormal when the v_j and the w_i are.

Guided proof 2.12 (Counting a product of two laws)**.** Let $\ell_i = i^{-\alpha}$ $(i \le m)$ and $r_j = j^{-\alpha}$ $(j \le n)$. By Warm-up 2.11 the eigenvalues of $R \otimes L$ are $(ij)^{-\alpha}$. Fix t, set $x = t^{-1/\alpha}$, and assume $x \le \min(m, n)$.

(a) Show $N(t) = \#\{(i,j) : ij \le x\} = \sum_{i \le x} \lfloor x/i \rfloor$.

(b) Using $x/i - 1 < \lfloor x/i \rfloor \le x/i$ and $H_n = \ln n + O(1)$, show $N(t) = x \ln x + O(x)$.

(c) (Optional: Dirichlet's hyperbola trick.) Every lattice point under the hyperbola $ij \le x$ has $i \le \sqrt{x}$ or $j \le \sqrt{x}$. Count the two strips and subtract the square counted twice: $N = 2 \sum_{i \le \sqrt{x}} \lfloor x/i \rfloor - \lfloor \sqrt{x} \rfloor^2$. Deduce $N = x \ln x + (2\gamma - 1)x + O(\sqrt{x})$.

(d) Invert. If $k = x \ln x$, show $x = k/\ln x$ and $\ln x = \ln k - \ln \ln x$. Hence $x = (k/\ln k)(1 + o(1))$ and $\lambda_k \approx (k/\ln k)^{-\alpha}$: a power law up to a logarithm.

(e) Using $k \approx x \ln x$, show that the local log–log slope is $d \ln \lambda / d \ln k = -\alpha/(1 + 1/\ln x)$, shallower than $-\alpha$.

Compute 2.13. Build the $64 \cdot 256$ eigenvalues $(ij)^{-1}$ of the product of two pure $\alpha = 1$ laws with $m = 64$ and $n = 256$, sort them, and fit a straight line to $\log \lambda_k$ against $\log k$ on ranks 17–200. You should find the exponent 0.77. Step (e), evaluated at the geometric mean rank k of the window with $x = k/\ln k$, predicts 0.75. (Reference: the end of `checks` in `make_figures.py`.)

Remark 2.14. Exact power laws in SOAP's two factors therefore appear shallower, and not exactly straight, in the product. A fitted exponent of a Kronecker-structured matrix should not be compared one-to-one with that of an unstructured matrix.

3 What the exponent decides

Many quantities that an optimizer cares about are sums over the spectrum: the trace, the Frobenius norm, $\operatorname{tr} A^{-1}$, the effective dimension under damping. For a power law each such sum is dominated either by the few largest eigenvalues (the *head*) or by the very many small ones (the *tail*), and the exponent decides which. This matters because the lab fits only the head. A fit decides head-dominated quantities and leaves tail-dominated ones open.

Reading. Graham, Knuth & Patashnik (1994), Section 6.3, on harmonic numbers, and Chapter 9 on asymptotics and Euler's summation formula, the precise version of Warm-up 3.1. Hastie, Tibshirani & Friedman (2009), Section 3.4.1, introduce the effective degrees of freedom of ridge regression, which is the effective dimension of Section 3.2.

3.1 Sums and integrals

Warm-up 3.1 (Sums against integrals)**.** Let $f \geq 0$ be decreasing on $[m, \infty)$. Draw rectangles of width 1 under and over the graph of f and show

$$\int_m^{M+1} f(x)\,dx \leq \sum_{k=m}^{M} f(k) \leq f(m) + \int_m^M f(x)\,dx.$$

Deduce $\ln(n+1) \leq H_n \leq 1 + \ln n$.

Guided proof 3.2 (Power sums in three regimes)**.** Let $S_s(K) = \sum_{k=1}^{K} k^{-s}$.

(a) ($s > 1$.) Show $\sum_{k>m} k^{-s} \leq m^{1-s}/(s-1)$. So the first m ranks carry all but a fraction of at most $m^{1-s}/((s-1)\zeta(s))$ of the full sum $\zeta(s)$. For $s = 2$ and $m = 16$ the bound is 0.038, and the true fraction is 0.037.

(b) ($s = 1$.) Show that each dyadic block of ranks $\{2^j, \ldots, 2^{j+1} - 1\}$ contributes $\ln 2 + O(2^{-j})$ to H_K: every block contributes about the same.

(c) ($0 \leq s < 1$.) Show $S_s(K) = K^{1-s}/(1-s) + O(1)$, and that the ranks above $K/2$ carry a fraction $1 - 2^{s-1} + o(1)$ of $S_s(K)$.

(d) ($s < 0$.) The terms now increase. Reverse the inequalities of Warm-up 3.1 and show $S_s(K) = K^{1-s}/(1-s)\,(1 + o(1))$.

Exercise 3.3 (Which part of a pure law decides a trace functional)**.** Let $\lambda_k = c\,k^{-\alpha}$ for $k = 1, \ldots, D$, so $\operatorname{tr} A^q = c^q S_{\alpha q}(D)$. Fill in the table. A functional is decided by the head when it converges as $D \to \infty$, so that a fixed number of top ranks carries almost all of it. Give the growth of $\operatorname{tr} A^{-1}$ with D.

Functional	decided by the head when	marginal at	decided by the tail when
$\operatorname{tr} A^2 = \|A\|_{\mathrm{F}}^2$			
$\operatorname{tr} A$			
$\operatorname{tr} A^{1/2}$			
$\operatorname{tr} A^{-1}$			

Which of these does a fit on ranks 1–200 decide when $\alpha \approx 1$? Keep in mind that preconditioning uses negative powers.

Exercise 3.4 (At $\alpha = 1$ every octave carries the same trace)**.** Let $\alpha = 1$.

(a) Show that the eigenvalues in a full octave $[t, 2t)$ carry trace $c \ln 2 + O(t)$.

(b) Show that the top-k share of the trace is $H_k/H_D \approx \ln k / \ln D$.

(c) Compute it for $k = 16$ and $D = 640$, 4,096 and 16,384. (Answers: 0.480, 0.380 and 0.329.)

Lab connection ([L])**.** In `spectral_rotation_20261020` (raw per-tensor GGN on one fixed batch), the late fits (steps 250–500) give $\alpha \approx 0.9$–1.1 on ranks 1–200, with a slightly flatter head ($\alpha_{\text{top}} = 0.65$–$0.86$) and $\alpha_{\text{bulk}} = 0.96$–$1.17$. The median top-16 share of the exact block trace over the 39 tensors is 0.35–0.45 at step 400 and 0.33–0.42 at step 500. A pure $\alpha = 1$ law over all ranks gives 0.380 for the 24 blocks with $D = 4{,}096$ and 0.329 for the 12 with $D = 16{,}384$. The trace is the only measured quantity that sees the whole spectrum, and it is consistent with the law extending well below rank 200. It is one number per tensor, so this is a consistency check, not a test.

Measured spectra also leave any power law at the bottom (finite data, symmetries). In `noise_scale_20261021` the smallest Ritz value of a damped whitened block $K + \delta I$ was $0.9987\,\delta$, so at least one block has eigenvalues at numerical zero. The tail-dominated rows of Exercise 3.3 are then set by the damping, not by the law.

3.2 Effective dimension under damping

A damped preconditioner $(A+\delta I)^{-1}$ treats directions with $\lambda \gg \delta$ as curved and directions with $\lambda \ll \delta$ as flat. The *effective dimension*

$$d_\delta(A) = \operatorname{tr} A(A+\delta I)^{-1} = \sum_k \frac{\lambda_k}{\lambda_k+\delta}$$

counts the curved directions softly: each contributes a number between 0 and 1. In ridge regression it is the effective degrees of freedom.

Warm-up 3.5. For $A \in \mathbb{S}_+^D$ and $\delta > 0$ show $\frac{1}{2}N_A(\delta) \le d_\delta(A) \le N_A(\delta) + \delta^{-1}\sum_{\lambda_k<\delta}\lambda_k$.

Guided proof 3.6 (Effective dimension of a power law)**.** Let $\lambda_k = c\,k^{-\alpha}$ and $a = \delta/c$.

(a) Let $f(x) = 1/(1+ax^\alpha)$. Show that $d_\delta = \sum_k f(k)$, that f is decreasing with $f(0) = 1$, and that for $D = \infty$ and $\alpha > 1$, $0 \le \int_0^\infty f - \sum_{k\ge1} f(k) \le 1$.

(b) Substitute $u = a^{1/\alpha}x$ to get $\int_0^\infty f = a^{-1/\alpha}\int_0^\infty du/(1+u^\alpha)$.

(c) Substitute $w = u^\alpha$ and use the Beta-function identity $\int_0^\infty w^{s-1}/(1+w)\,dw = \pi/\sin(\pi s)$ for $0 < s < 1$ (a Fact; see any text on the Gamma function) to get $\int_0^\infty du/(1+u^\alpha) = (\pi/\alpha)/\sin(\pi/\alpha)$. Check the case $\alpha = 2$ by hand.

(d) Conclude $\left|d_\delta - (c/\delta)^{1/\alpha}(\pi/\alpha)/\sin(\pi/\alpha)\right| \le 1$. Damping truncates the law near rank $N_A(\delta) \approx (c/\delta)^{1/\alpha}$, with a factor $(\pi/\alpha)/\sin(\pi/\alpha) > 1$ from the soft edge.

(e) ($\alpha = 1$, finite D.) Show $\left|d_\delta - (c/\delta)\ln(1+D\delta/c)\right| \le 1$. Here the directions below δ add a logarithmic factor.

Compute 3.7. Check (d) for $\alpha = 2$ and $\delta = 10^{-6}c$, summing to rank 10^7: 1570.2 against 1570.8. Check (e) for $\alpha = 1$, $D = 4{,}096$ and $\delta = 10^{-3}c$: 1628.1 against 1628.5. The same damping keeps very different numbers of directions active for different exponents.

4 The spectrum as a probability distribution

Suppose you know the spectrum of A exactly but nothing about its frame. What should you believe about the curvature $v^\top Av$ along a direction v you care about? The natural model puts the uniform distribution on the frame. You will show that the curvature along v is then an average of the eigenvalues with random weights, that the weights have mean $1/D$, and that the average concentrates around $\operatorname{tr} A/D$ at a rate set by the spread of the spectrum. This is the precise sense in which "the spectrum is a distribution". The second half of the section asks what changes once we compute: a matrix–vector product reveals the spectrum *as seen from* a vector.

> **Reading.** Meckes (2019), Chapter 1, defines Haar measure on $\mathrm{O}(D)$ and gives several constructions; Mezzadri (2007) shows how to sample it correctly (QR of a Gaussian matrix with a sign fix). Vershynin (2018), Chapter 3, covers Gaussian vectors, the uniform distribution on the sphere and concentration of the norm. Gelman et al. (2013), Appendix A, lists the Gamma and Dirichlet laws. Trefethen & Bau (1997), Lectures 36–38, derive Lanczos, its link to Gauss quadrature and conjugate gradients; Golub & Meurant (2010) is the full treatment. Hennig, Osborne & Kersting (2022), the part on linear algebra, reads these methods as inference.

4.1 Spectrum known, frame unknown: the Haar model

Fact 4.1 (Haar measure)**.** *There is a unique probability distribution on* $\mathrm{O}(D)$*, the* Haar *distribution, that is invariant under multiplication: if* Q *is Haar distributed and* $R \in \mathrm{O}(D)$ *is fixed, then* RQ *and* QR *are Haar distributed too.*

Fact 4.2 (Direction and length of a Gaussian vector)**.** *Let $g \sim \mathcal{N}(0, I_D)$. Then $g/\|g\|$ is uniformly distributed on the unit sphere and independent of $\|g\|$.*

Definition 4.3 (Haar model)**.** Fix Λ, draw Q from the Haar distribution, and set $A = Q\Lambda Q^\top$. This models a matrix whose spectrum is known and whose frame is not. For a fixed unit vector v let $w_k = (q_k^\top v)^2$, with q_k the columns of Q.

Warm-up 4.4. (a) If $g \sim \mathcal{N}(0, I_D)$ and $R \in \mathrm{O}(D)$, then $Rg \sim \mathcal{N}(0, I_D)$.

(b) In the Haar model, $Q^\top v$ is uniformly distributed on the unit sphere. (Use Fact 4.1.)

(c) $\sum_k w_k = 1$, and $\mathbb{E}w_k = 1/D$ for every k.

Guided proof 4.5 (The curvature along a direction when the frame is unknown)**.** Let $\psi = Q^\top v$, so that $w_k = \psi_k^2$.

(a) Show $v^\top f(A)v = \sum_k f(\lambda_k)\, w_k$ for every f.

(b) Write $\psi = g/\|g\|$ with $g \sim \mathcal{N}(0, I_D)$ (Warm-up 4.4 and Fact 4.2). Use the independence in Fact 4.2 to show $\mathbb{E}[g_k^4] = \mathbb{E}[\|g\|^4]\,\mathbb{E}[w_k^2]$ and $\mathbb{E}[g_k^2 g_l^2] = \mathbb{E}[\|g\|^4]\,\mathbb{E}[w_k w_l]$.

(c) Using $\mathbb{E}g_k^4 = 3$, show $\mathbb{E}\|g\|^4 = D(D+2)$. Deduce $\mathbb{E}w_k^2 = 3/(D(D+2))$ and, for $k \neq l$, $\mathbb{E}w_k w_l = 1/(D(D+2))$.

(d) Deduce $\operatorname{Var} w_k = 2(D-1)/(D^2(D+2))$ and $\operatorname{Cov}(w_k, w_l) = -2/(D^2(D+2))$ for $k \neq l$.

(e) Show $\mathbb{E}[v^\top f(A)v] = \int f\, d\mu_A$ and

$$\operatorname{Var}[v^\top f(A)v] = \frac{2\operatorname{Var}_{\mu_A}(f)}{D+2}, \qquad \operatorname{Var}_{\mu_A}(f) = \frac{1}{D}\sum_k f(\lambda_k)^2 - \Big(\frac{1}{D}\sum_k f(\lambda_k)\Big)^2.$$

(f) The relative standard deviation of the Rayleigh quotient $v^\top A v$ is therefore $\sqrt{2/(D+2)}$ times the coefficient of variation of μ_A. Compute it for a pure $\alpha = 1$ law with $D = 4{,}096$. (Answer: 0.203.)

Exercise 4.6 (The Dirichlet law (optional))**.** (a) Show that g_k^2 has the χ_1^2 law, which is the Gamma law with shape $\frac{1}{2}$ and scale 2.

(b) Use the fact that independent Gamma variables with a common scale, divided by their sum, are Dirichlet distributed (Gelman et al. 2013, Appendix A) to show $(w_1, \dots, w_D) \sim \text{Dirichlet}(\frac{1}{2}, \dots, \frac{1}{2})$.

(c) The Dirichlet$(a_1, \dots, a_D)$ moments are $\mathbb{E}w_k = a_k/a_0$, $\operatorname{Var} w_k = a_k(a_0 - a_k)/(a_0^2(a_0+1))$ and $\operatorname{Cov}(w_k, w_l) = -a_k a_l/(a_0^2(a_0+1))$, with $a_0 = \sum_k a_k$. Check them against Guided proof 4.5(d).

Exercise 4.7 (The spectrum as a predictive distribution)**.** In the Haar model, draw an index I with $\Pr(I = k \mid Q) = w_k$. Show that $\Pr(I = k) = 1/D$, so $\lambda_I \sim \mu_A$, and that $v^\top f(A)v = \mathbb{E}[f(\lambda_I) \mid Q]$.

Compute 4.8. Check Guided proof 4.54.5(e) for $f(\lambda) = \lambda$ by Monte Carlo, with a pure $\alpha = 1$ law and $D = 256$. You need not sample Q: by Warm-up 4.4, sample $\psi = g/\|g\|$ directly. With 200,000 draws, the ratio of the empirical to the predicted variance was 1.0016 and the ratio of the means 0.9995. (Reference: `checks` in `make_figures.py`.)

Remark 4.9 (The probabilistic-numerics reading)*.* Exercise 4.7 is the precise sense of "the spectrum is a distribution". When nothing is known about the frame, μ_A is the prior predictive distribution of the curvature that any direction will meet. A matrix–vector product Av is an observation that changes this: it reveals part of the spectral measure seen from v (Section 4.2). In the language of probabilistic numerics (Hennig 2015; Wenger & Hennig 2020; Hennig, Osborne & Kersting 2022), the spectrum plays the role of the prior, and computation (the products) is inference about the frame relative to the vectors we care about. Part III develops this.

4.2 The spectral measure seen from a vector

Definition 4.10. For $v \in \mathbb{R}^D$, the *spectral measure of A seen from v* is $\mu_{A,v} = \sum_k (u_k^\top v)^2 \, \delta_{\lambda_k}$. Its total mass is $\|v\|^2$; its normalisation is $\nu_{A,v} = \mu_{A,v}/\|v\|^2$.

Warm-up 4.11. (a) Show $\int f \, d\mu_{A,v} = v^\top f(A) v$ for every f, and that the total mass is $\|v\|^2$.

(b) Show $\mu_{QAQ^\top, Qv} = \mu_{A,v}$ for all $Q \in \mathrm{O}(D)$.

(c) In the Haar model, show $\mathbb{E}\,\mu_{A,v} = \|v\|^2 \mu_A$ (as measures: the expected weight of each atom).

(d) Let $A = \mathrm{diag}(\lambda_1, \dots, \lambda_D)$ with distinct entries. Compute $\nu_{A,v}$ for $v = e_1$ and for $v = (1, \dots, 1)/\sqrt{D}$. The same matrix looks entirely different from different vectors.

4.3 Krylov methods read only the measure seen from the start vector

Fact 4.12 (Lanczos)**.** *Let v_1 be a unit vector, and assume that the Krylov space*

$$\mathcal{K}_m = \mathrm{span}\{v_1, Av_1, \dots, A^{m-1}v_1\}$$

has dimension m. The Lanczos iteration builds an orthonormal basis $V_m = [v_1, \dots, v_m]$ of $\mathcal{K}_m$ and a symmetric tridiagonal $T_m = V_m^\top A V_m$ such that

$$AV_m = V_m T_m + \beta_m v_{m+1} e_m^\top,$$

with $v_{m+1} \perp \mathcal{K}_m$ a unit vector and e_m the last unit vector of $\mathbb{R}^m$.

Fact 4.13 (Conjugate gradients)**.** *For $A \succ 0$, the CG iterate x_m for $Ax = b$, started at $x_0 = 0$, minimises $\|x - y\|_A$ over $y \in \mathcal{K}_m(A, b)$, where $\|z\|_A^2 = z^\top A z$.*

Guided proof 4.14 (Lanczos computes a Gauss quadrature rule)**.** (a) Show that $T_m^j e_1$ has non-zero entries only in its first $j+1$ positions.

(b) Show by induction that $A^j v_1 = V_m T_m^j e_1$ for $0 \le j \le m-1$. Where do you need $j \le m-1$?

(c) Show $v_1^\top A^i v_1 = e_1^\top T_m^i e_1$ for $0 \le i \le 2m-1$. (Treat even $i = 2j$ and odd $i = 2j+1$ separately.)

(d) Write $T_m = S \Theta S^\top$ with nodes $\Theta = \mathrm{diag}(\theta_1, \dots, \theta_m)$ and weights $\tau_j = S_{1j}^2$. Show $\sum_j \tau_j \theta_j^i = \int \lambda^i \, d\nu_{A,v_1}(\lambda)$ for $0 \le i \le 2m-1$. The m-point measure $\sum_j \tau_j \delta_{\theta_j}$ matches $2m$ moments of ν_{A,v_1}: this is the m-point Gauss quadrature rule of ν_{A,v_1}.

(e) Interpret: m matrix–vector products reveal $2m$ moments of the spectrum seen from v_1. Show that T_m depends on A only through these moments, so two matrices that agree on them give the same output, however different their frames are elsewhere.

Guided proof 4.15 (The CG error is a spectral integral)**.** Let $A \succ 0$, $x = A^{-1}b$, and x_m the CG iterate.

(a) Write any $y \in \mathcal{K}_m(A, b)$ as $y = q(A)b$ with $\deg q < m$. Show $x - y = p(A)x$ with $p(\lambda) = 1 - \lambda q(\lambda)$, so $\deg p \le m$ and $p(0) = 1$. Conversely, every such p arises from some q.

(b) Show $\|p(A)x\|_A^2 = b^\top A^{-1} p(A)^2 b = \int p(\lambda)^2 \lambda^{-1} \, d\mu_{A,b}(\lambda)$.

(c) Conclude

$$\|x - x_m\|_A^2 = \min\Big\{ \int p(\lambda)^2 \lambda^{-1} \, d\mu_{A,b}(\lambda) : \deg p \le m, \; p(0) = 1 \Big\}.$$

Two matrices with different frames but the same measure seen from b produce the same CG errors at every step.

(d) If $\mu_{A,b}$ has only r atoms, show that CG is exact after r steps.

The `weights` returned by the lab's `noise_scale_20261021/scripts/lanczos.py` are exactly the τ_j of Guided proof 4.14(d).

4.4 One step on a quadratic, and the value of curvature

Exercise 4.16 (The best step along a direction)**.** Let $A \succ 0$, $g \in \mathbb{R}^D$ and $q(x) = \frac{1}{2}x^\top Ax - g^\top x$. For $f > 0$ on the spectrum, show that the best decrease along $x = \eta f(A)g$, $\eta \in \mathbb{R}$, is

$$\max_\eta [q(0) - q(\eta f(A)g)] = \frac{\left(\int f \, d\mu_{A,g}\right)^2}{2 \int \lambda f(\lambda)^2 \, d\mu_{A,g}(\lambda)}.$$

Guided proof 4.17 (Kantorovich's inequality)**.** Let ν be a probability measure on $[m, M]$ with $0 < m < M$, and $\kappa = M/m$.

(a) Use Exercise 4.16 to show that the ratio of the best decreases of Newton ($f = 1/\lambda$) and of a tuned gradient step ($f = 1$) is $\mathbb{E}_\nu[\lambda]\,\mathbb{E}_\nu[\lambda^{-1}]$ with $\nu = \nu_{A,g}$.

(b) Show $\mathbb{E}_\nu[\lambda]\,\mathbb{E}_\nu[\lambda^{-1}] \geq 1$ (Cauchy–Schwarz or Jensen).

(c) For $\lambda \in [m, M]$, show $(\lambda - m)(M - \lambda) \geq 0$ and deduce $\lambda + mM/\lambda \leq m + M$.

(d) Integrate against ν and use $2\sqrt{xy} \leq x + y$ to show $\mathbb{E}_\nu[\lambda]\,\mathbb{E}_\nu[\lambda^{-1}] \leq (m+M)^2/(4mM) = (\kappa+1)^2/(4\kappa)$.

Exercise 4.18 (Uniform weights on a power law)**.** Let $\lambda_k = c/k$, $k = 1, \ldots, D$, and $\nu = \mu_A$ (a gradient with equal weight on every eigen-direction). Show $\mathbb{E}_\mu[\lambda]\,\mathbb{E}_\mu[\lambda^{-1}] = H_D(D+1)/(2D) \approx \frac{1}{2}\ln D$. Compute it for $D = 4{,}096$ and $16{,}384$ (answers: 4.45 and 5.14) and compare with Kantorovich's worst case for $\kappa = D = 4{,}096$, which is 1,025. What decides the value of curvature: the spectrum or the weights?

Remark 4.19 (Two measures decide a step). With batch noise of covariance Σ, the denominator in Exercise 4.16 gains the term $(1/B - 1/N)\int \lambda f^2 \, d\mu_{A,\Sigma}$, where $\mu_{A,\Sigma} = \sum_k (u_k^\top \Sigma u_k)\, \delta_{\lambda_k}$ (`batch_noise.md`, T1). For one step, all that matters about A is the pair $(\mu_{A,\bar{g}}, \mu_{A,\Sigma})$: how the signal and the noise are spread over the spectrum. The frame enters only through these weights. Two questions follow for training. Does the gradient's measure $\nu_{A,\bar{g}}$ stay stationary while the frame rotates, that is, does the gradient co-rotate with the frame? And how do it and $\mu_{A,\Sigma}$ change during training? The forms $z^\top A^q z$, $q \in [-1, 1]$, measured in `noise_scale_20261021` are moments of exactly these measures.

5 Perturbation: what a power law implies about the frame

Curvature matrices in training are never known exactly. A different batch, a later step or a cheaper estimator gives $B = A + E$ instead of A. This section asks how far eigenvalues and eigenvectors move. The answer is asymmetric. Eigenvalues move by at most the size of E, whatever the spectrum. Eigenvectors move by the size of E divided by the gap to the neighbouring eigenvalues. A power law has gaps that shrink like λ_k/k, so going down the spectrum, eigenvectors are lost much earlier than eigenvalues.

Reading. Bhatia (1997) is the best single source: Chapter III (variational principles, Weyl), Chapter VI (Hoffman–Wielandt), Chapter VII (Davis–Kahan $\sin\Theta$ theorems) and Chapter X (perturbation of matrix functions). Horn & Johnson (2013), Sections 4.2–4.3, cover Courant–Fischer, Weyl and Schur's majorization. Yu, Wang & Samworth (2015) is a short paper written for statisticians that states the Davis–Kahan version we use; Davis & Kahan (1970) is the original, and Stewart & Sun (1990) the standard monograph. Kato (1976), Chapter II, treats analytic perturbation in finite dimensions.

Warm-up 5.1 (Everything in a 2×2 matrix)**.** Let $A = \operatorname{diag}(a, b)$ with gap $g = a - b > 0$, and $E = \delta \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ with $\delta > 0$.

(a) Show that the eigenvalues of $A+E$ are $\frac{a+b}{2} \pm \sqrt{g^2/4 + \delta^2}$. Each moves by $\sqrt{g^2/4 + \delta^2} - g/2$, which lies in $[0, \delta]$ and is about δ^2/g when $\delta \ll g$.

(b) Show that the top eigenvector of $A + E$ is $(\cos\theta, \sin\theta)$ with $\tan 2\theta = 2\delta/g$.

(c) For $a = 1$, $b = 0.99$ and $\delta = 0.01$, compute both. (Answers: the eigenvalues move by 0.0062, the eigenvector rotates by 31.7°.)

(d) Let $g \to 0$ with δ fixed. What happens to θ, and what does that say about who decides the eigenvector, A or E?

5.1 Eigenvalues are stable

Guided proof 5.2 (Courant–Fischer)**.** Show $\lambda_k(A) = \max_{\dim S = k} \min_{v \in S,\, \|v\| = 1} v^\top A v$, the maximum over subspaces S of dimension k.

(a) For $S = \operatorname{span}(u_1, \dots, u_k)$, show that the inner minimum is λ_k.

(b) For any S of dimension k, show $S \cap \operatorname{span}(u_k, \dots, u_D) \neq \{0\}$ by counting dimensions.

(c) For a unit v in that intersection show $v^\top A v \leq \lambda_k$, and conclude.

Guided proof 5.3 (Weyl's inequality)**.** Let $B = A + E$. Show $|\lambda_k(B) - \lambda_k(A)| \leq \|E\|_{\mathrm{op}}$ for every k.

(a) Show $v^\top B v \leq v^\top A v + \|E\|_{\mathrm{op}}$ for every unit v.

(b) Insert this into Courant–Fischer to get $\lambda_k(B) \leq \lambda_k(A) + \|E\|_{\mathrm{op}}$. Swap the roles of A and B.

Fact 5.4 (Birkhoff–von Neumann)**.** *A matrix with non-negative entries whose rows and columns all sum to 1 is* doubly stochastic*. Every doubly stochastic matrix is a convex combination of permutation matrices. Hence a linear function on the doubly stochastic matrices attains its maximum at a permutation matrix. (Bhatia 1997, Chapter II.)*

Exercise 5.5 (Rearrangement inequality)**.** Let $a_1 \geq \cdots \geq a_D$ and $b_1 \geq \cdots \geq b_D$. Show $\sum_i a_i b_{\pi(i)} \leq \sum_i a_i b_i$ for every permutation π. Start with two indices: if $i < j$ but $\pi(i) > \pi(j)$, swapping the two values does not decrease the sum.

Guided proof 5.6 (Hoffman–Wielandt)**.** Let $A = U_A \operatorname{diag}(a) U_A^\top$ and $B = U_B \operatorname{diag}(b) U_B^\top$ with sorted spectra a and b. Show $\sum_k (a_k - b_k)^2 \leq \|A - B\|_{\mathrm{F}}^2$.

(a) Show $\|A - B\|_{\mathrm{F}}^2 = \sum_i a_i^2 + \sum_j b_j^2 - 2\operatorname{tr}(AB)$.

(b) With $W = U_A^\top U_B$, show $\operatorname{tr}(AB) = \sum_{i,j} a_i b_j W_{ij}^2$.

(c) Show that the matrix $P = (W_{ij}^2)$ is doubly stochastic.

(d) Use Fact 5.4 and Exercise 5.5 to show $\operatorname{tr}(AB) \leq \sum_i a_i b_i$, and conclude.

Exercise 5.7 (Spectra are close as distributions)**.** For two probability measures on $\mathbb{R}$ that put mass $1/D$ on each of D points, an optimal coupling for the Wasserstein-2 distance can be taken to be a permutation (Fact 5.4 again; Santambrogio 2015, Chapter 2). Use Exercise 5.5 to show that the sorted coupling is optimal, and deduce $W_2(\mu_A, \mu_B) \leq \|A - B\|_{\mathrm{F}}/\sqrt{D}$.

5.2 Eigenvectors are only as stable as their gaps

Guided proof 5.8 (A one-vector $\sin\theta$ theorem)**.** Let $B = A + E$, fix k, and let $g_k = \min_{j\neq k}|\lambda_j(A) - \lambda_k(A)| > 0$ be the gap of $\lambda_k(A)$ to the rest of the spectrum. Let $u = u_k(A)$, $v = u_k(B)$, $\mu = \lambda_k(B)$, and $P = I - uu^\top$. Then $\sin\angle(u, v) = \|Pv\|$.

(a) Show $(A - \mu I)v = -Ev$.

(b) Show that P commutes with A, so $P(A - \mu I)v = (A - \mu I)Pv$.

(c) Pv lies in $\operatorname{span}\{u_j : j \neq k\}$. Show that on this span $\|(A - \mu I)z\| \geq \min_{j\neq k}|\lambda_j - \mu|\,\|z\|$, and use Weyl to get $\min_{j\neq k}|\lambda_j - \mu| \geq g_k - \|E\|_{\mathrm{op}}$.

(d) Combine (a)–(c) with $\|PEv\| \leq \|E\|_{\mathrm{op}}$ to get $\sin\angle(u, v) \leq \|E\|_{\mathrm{op}}/(g_k - \|E\|_{\mathrm{op}})$ when $\|E\|_{\mathrm{op}} < g_k$.

(e) Conclude $\sin\angle(u, v) \leq 2\|E\|_{\mathrm{op}}/g_k$ in all cases.

Remark 5.9 (The subspace version). The same idea, with P replaced by the projector onto the complement of the top-k eigenspace, gives the version of Yu, Wang & Samworth (2015) for the top-k subspaces: $\|\sin\Theta\|_{\mathrm{F}} \leq 2\min(\sqrt{k}\|E\|_{\mathrm{op}}, \|E\|_{\mathrm{F}})/(\lambda_k - \lambda_{k+1})$, where $\sin\Theta$ collects the sines of the principal angles between the two subspaces. Only the gap at the cut enters, not the gaps inside the subspace.

Exercise 5.10 (Gaps of a power law)**.** Let $\lambda_k = c\,k^{-\alpha}$.

(a) Show that $h(x) = 1 - (1 + x)^{-\alpha}$ is concave with $h(0) = 0$, and deduce $xh'(x) \leq h(x) \leq xh'(0)$ for $x \geq 0$.

(b) With $x = 1/k$, deduce $\frac{\alpha}{k}(1 + \frac{1}{k})^{-\alpha-1} \leq (\lambda_k - \lambda_{k+1})/\lambda_k \leq \frac{\alpha}{k}$.

(c) Show that consecutive gaps decrease, so that for $2 \leq k < D$ the gap g_k of Guided proof 5.8 is $\lambda_k - \lambda_{k+1}$ (and $g_1 = \lambda_1 - \lambda_2$).

Guided proof 5.11 (Two resolution limits)**.** Let A follow a pure law (c, α), let $\|B - A\|_{\mathrm{op}} \leq \eta$, and let $s \in (0, 1]$.

(a) Show $|\lambda_k(B) - \lambda_k(A)| \leq s\lambda_k(A)$ for every $k \leq K_{\mathrm{val}} := (sc/\eta)^{1/\alpha}$.

(b) Use Exercise 5.10 and $1 + 1/k \leq 2$ to show $g_k \geq \alpha c\,k^{-\alpha-1}2^{-\alpha-1}$ for $k < D$.

(c) Use Guided proof 5.8 to show $\sin\angle(u_k(A), u_k(B)) \leq s$ for every $k < D$ with $k \leq K_{\mathrm{vec}} := \big(s\alpha c/(2^{\alpha+2}\eta)\big)^{1/(\alpha+1)}$.

(d) Show $K_{\mathrm{vec}}/K_{\mathrm{val}} \to 0$ as $\eta \to 0$: for small perturbations far more eigenvalues than eigenvectors are resolved. Eigenvalues are resolved to depth $\propto \eta^{-1/\alpha}$, eigenvectors only to $\propto \eta^{-1/(\alpha+1)}$.

Exercise 5.12 (Numbers for the guarantees)**.** Take $\alpha = 1$, $c = 1$, $s = 1/2$ and $\eta = 8.9 \times 10^{-4}$, the operator norm of the Gaussian perturbation used in Figure 2. Compute K_{val} and K_{vec}. (Answers: 560 and 8.4.) The simulation in Figure 2 resolves eigenvalues to rank 1,801 and eigenvectors to rank 169. Which guarantee is more pessimistic, and why might a worst-case bound be pessimistic for a random E?

5.3 Typical perturbations: first-order theory

Worst-case bounds assume that E is aligned as badly as possible with the frame of A. A random E is not. First-order perturbation theory gives the typical sizes with explicit constants.

Guided proof 5.13 (First-order perturbation theory)**.** Let A have distinct eigenvalues and $A(\varepsilon) = A + \varepsilon E$. Accept (Kato 1976, Chapter II) that $\lambda_k(\varepsilon)$ and unit eigenvectors $u_k(\varepsilon)$ can be chosen differentiable near $\varepsilon = 0$. Write $'$ for $d/d\varepsilon$ at $\varepsilon = 0$.

(a) Differentiate $A(\varepsilon)u_k(\varepsilon) = \lambda_k(\varepsilon)u_k(\varepsilon)$ to get $Eu_k + Au_k' = \lambda_k' u_k + \lambda_k u_k'$.

(b) Multiply by $u_k^\top$ to get $\lambda_k' = u_k^\top E u_k$.

(c) Multiply by $u_j^\top$, $j \neq k$, to get $u_j^\top u_k' = u_j^\top E u_k/(\lambda_k - \lambda_j)$.

(d) Differentiate $u_k^\top u_k = 1$ to get $u_k^\top u_k' = 0$. Conclude

$$u_k' = \sum_{j \neq k} \frac{u_j^\top E u_k}{\lambda_k - \lambda_j}\, u_j, \qquad \sin\angle(u_k(0), u_k(\varepsilon)) = \varepsilon\|u_k'\| + O(\varepsilon^2).$$

Compare with Guided proof 1.81.8(d).

Exercise 5.14 (Typical sizes for Gaussian noise)**.** Let $E = \varepsilon G$, where G is symmetric with independent centred Gaussian entries, of variance 1 off the diagonal and 2 on it. Then $Q^\top G Q$ has the law of G for every fixed $Q \in \mathrm{O}(D)$ (accept this), and $\|E\|_{\mathrm{op}} \approx 2\varepsilon\sqrt{D}$.

(a) Show that $u_j^\top G u_k$, $j \neq k$, are independent $\mathcal{N}(0,1)$ and $u_k^\top G u_k \sim \mathcal{N}(0,2)$. Deduce that, to first order, $\mathbb{E}\sin^2\angle(u_k(A), u_k(B)) \approx \varepsilon^2 \sum_{j\neq k}(\lambda_k - \lambda_j)^{-2}$, and that an eigenvalue moves by about $\sqrt{2}\varepsilon$.

(b) For a pure law, split the sum, and use $g_k \approx \alpha c\, k^{-\alpha-1}$. Show that the neighbouring ranks contribute $\sum_{m\neq 0}(m g_k)^{-2} \approx 2\zeta(2)k^{2\alpha+2}/(\alpha^2 c^2)$, and that the many much smaller eigenvalues contribute about $Dk^{2\alpha}/c^2$.

(c) Show that the neighbours dominate once $k \gtrsim \alpha\sqrt{D/(2\zeta(2))}$. For $\alpha = 1$ and $D = 2{,}000$ this is about 25.

(d) Show that the typical eigenvector resolution limit, where the sine reaches s, is $k_{\mathrm{vec}} \approx \left(s\alpha c/(\sqrt{2\zeta(2)}\,\varepsilon)\right)^{1/(\alpha+1)}$. For $\alpha = 1$, $c = 1$, $s = 1/2$ and $\varepsilon = 10^{-5}$ this is 166.

Compute 5.15. Reproduce Figure 2(a) and (b). Take $\lambda_k = 1/k$ with $D = 2{,}000$, $E = \varepsilon(Z + Z^\top)/\sqrt{2}$ with Z a standard Gaussian matrix and $\varepsilon = 10^{-5}$, and compute the eigendecomposition of $\mathrm{diag}(\lambda) + E$. Why may you take A diagonal? Over three draws (medians) you should find relative eigenvalue errors below 3.4×10^{-3} in the top 100 ranks, reaching 0.5 only at rank 1,801, and eigenvector sines reaching 0.5 at rank 169 (predicted 166). (Reference: `figure_resolution` in `make_figures.py`.)

5.4 Functions of the matrix: hard and soft cuts

A preconditioner is a function of an estimated matrix. Which functions survive a perturbation? Smooth ones do, in the Frobenius norm, whatever the eigenvectors did. A hard cut onto the top k eigenvectors survives only as well as the gap at the cut.

Guided proof 5.16 (Lipschitz spectral functions are as stable as the matrix)**.** Let f be L-Lipschitz on an interval that contains the spectra of $A = U_A\,\mathrm{diag}(a)U_A^\top$ and $B = U_B\,\mathrm{diag}(b)U_B^\top$. Show $\|f(A) - f(B)\|_{\mathrm{F}} \leq L\|A - B\|_{\mathrm{F}}$.

(a) With $W = U_A^\top U_B$, show that $U_A^\top(A - B)U_B$ has entries $(a_i - b_j)W_{ij}$.

(b) Show that $U_A^\top(f(A) - f(B))U_B$ has entries $(f(a_i) - f(b_j))W_{ij}$.

(c) Use that the Frobenius norm is unchanged by orthogonal factors, and compare the two matrices entry by entry.

In the operator norm the same statement fails with constant L; see Bhatia (1997), Chapter X.

Exercise 5.17 (Damping sets the stability of a fractional step)**.** Show that $f(\lambda) = (\lambda + \delta)^{-\nu}$, $\nu > 0$, is $\nu\delta^{-\nu-1}$-Lipschitz on $[0, \infty)$. Deduce $\|(A + \delta I)^{-\nu} - (B + \delta I)^{-\nu}\|_{\mathrm{F}} \leq \nu\delta^{-\nu-1}\|A - B\|_{\mathrm{F}}$ for PSD A and B, whatever the eigenvectors did.

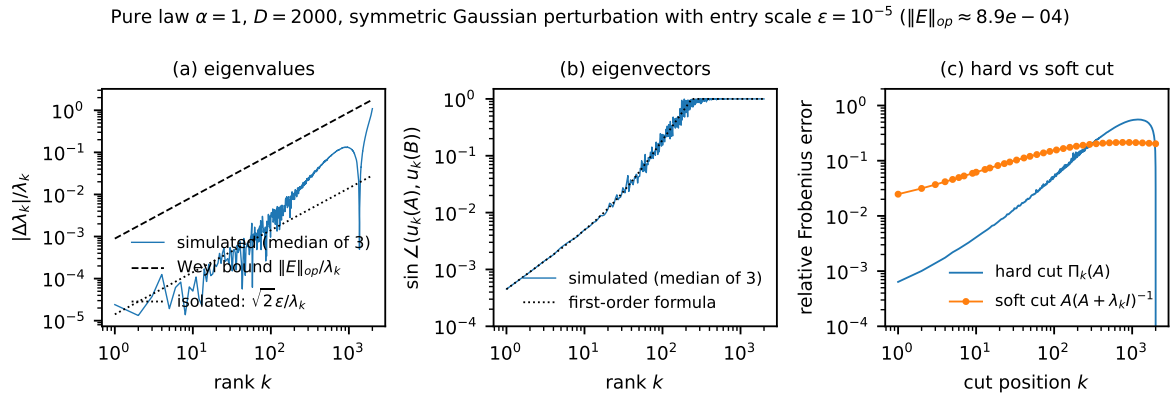


Figure 2: Resolution limits (synthetic). A pure $\alpha = 1$ law, $D = 2{,}000$, perturbed by a symmetric Gaussian matrix with entry scale $\varepsilon = 10^{-5}$; medians of three draws. (a) Relative eigenvalue errors, with Weyl's bound and the isolated-eigenvalue estimate $\sqrt{2}\varepsilon/\lambda_k$. (b) Eigenvector angles, with the first-order formula of Exercise 5.14. (c) Relative Frobenius error of the top-k projector (hard cut) and of the soft cut $A(A + \lambda_k I)^{-1}$ at the same place (Remark 5.19).

Exercise 5.18 (Distance between projectors)**.** Let P and P' be orthogonal projectors of rank k, and let $\cos^2\theta_1, \dots, \cos^2\theta_k$ be the squared singular values of $V^\top V'$ for orthonormal bases V and V' of their ranges. Show $\|P - P'\|_{\mathrm{F}}^2 = 2(k - \|PP'\|_{\mathrm{F}}^2) = 2\sum_i \sin^2\theta_i = 2\|\sin\Theta\|_{\mathrm{F}}^2$. With Remark 5.9, the top-k projector Π_k satisfies $\|\Pi_k(A) - \Pi_k(B)\|_{\mathrm{F}} \le 2\sqrt{2}\min(\sqrt{k}\|E\|_{\mathrm{op}}, \|E\|_{\mathrm{F}})/(\lambda_k - \lambda_{k+1})$: no bound of the kind in Guided proof 5.16 holds, because the indicator of $\{\lambda \ge \lambda_k\}$ is not Lipschitz.

Remark 5.19 (A cut or a smooth function?). The two bounds have different strengths, and Figure 2c shows both effects. A hard cut where the gap is large is far more stable than a smooth "soft cut" $A(A+\lambda_k I)^{-1}$ at the same place: at $k = 16$ the relative errors are 0.007 (hard) and 0.0739 (soft). The smooth function acts on every direction and pays for the perturbation everywhere; its error is controlled by $\|E\|_{\mathrm{F}}$, which grows like εD. Deep in the spectrum, where gaps are tiny, the order reverses: at $k \approx 1{,}000$ the errors are 0.54 (hard) and 0.214 (soft). The curves cross at $k \approx 346$, about twice the eigenvector resolution limit. (Soft-cut values are on a grid of cut positions; the ones quoted are at 15, 193 and 917.)

The design rule this suggests is not "never store a frame" but: cut only where the frame is resolved, and use smooth functions of the current matrix beyond that. PARSEC-L's brake $I - \sum_{k \le r} \frac{\kappa_k}{1+\kappa_k} w_k w_k^\top$ with $\kappa_k = c_{\mathrm{b}}\eta\lambda_k$ is exactly the soft function $\lambda/(\lambda + 1/(c_{\mathrm{b}}\eta))$ applied inside a hard top-r cut. Its fully smooth version is $(I + c_{\mathrm{b}}\eta K)^{-1}$, which needs no eigenvectors.

5.5 Stale frame, fresh diagonal

SOAP and EKFAC keep a frame U computed from an earlier matrix and refresh only a diagonal in that frame. What is the best diagonal for a given frame, and what does a stale frame cost?

Warm-up 5.20 (Pinching)**.** For a frame U and $A \in \mathbb{S}^D$ let

$$\mathcal{P}_U(A) = U\,\mathrm{diag}(u_1^\top A u_1, \dots, u_D^\top A u_D)\,U^\top.$$

Among the matrices $UMU^\top$ with M diagonal, show that $\mathcal{P}_U(A)$ is the closest to A in Frobenius norm, and that $\|A - \mathcal{P}_U(A)\|_{\mathrm{F}}^2 = \sum_{i \ne j}(u_i^\top A u_j)^2$. Show also $\mathrm{tr}\,\mathcal{P}_U(A) = \mathrm{tr}\,A$.

Guided proof 5.21 (Ky Fan's principle and Schur's majorization)**.** (a) Let $x_1, \dots, x_k$ be orthonormal and $X = [x_1, \dots, x_k]$. Show $\sum_i x_i^\top A x_i = \sum_j \lambda_j c_j$ with $c_j = \|X^\top u_j\|^2$.

(b) Show $0 \le c_j \le 1$ and $\sum_j c_j = k$.

(c) Show that $\sum_j \lambda_j c_j$ under these constraints is at most $\lambda_1 + \cdots + \lambda_k$.

(d) Apply this to the k columns of U with the largest Rayleigh quotients $u_i^\top A u_i$. Conclude that the Rayleigh quotients are *majorised* by the eigenvalues: for every k the sum of the k largest is at most $\lambda_1 + \cdots + \lambda_k$, with equality for $k = D$.

(e) Deduce $\max_i u_i^\top A u_i \le \lambda_1(A)$ and $\min_i u_i^\top A u_i \ge \lambda_D(A)$: a stale frame flattens the effective spectrum.

Remark 5.22 (The bridge to SOAP). This is the structure of EKFAC and SOAP: a frame refreshed rarely and a diagonal refreshed every step. Warm-up 5.20 is the Frobenius-optimality argument that George et al. (2018) give for EKFAC's diagonal, and Eschenhagen et al. (2025) analyse Shampoo and SOAP in the same Frobenius terms. Guided proof 5.21 adds what a stale frame costs spectrally: the effective spectrum is flattened, with the top lowered and the bottom raised, the more so the further the frame has drifted. That is the direction of the shrinkage that rotationally invariant estimators from random-matrix statistics apply when eigenvectors are unreliable (Bun, Bouchaud & Potters 2017). Two cautions for Part III: SOAP's diagonal is an exponential average of squared rotated gradients, a second moment of the gradient rather than the GGN; and it enters through a square root.

> **Lab connection** ([L]). Late in training, the 10-step capture of a cached top-k subspace rises with k: 0.78–0.84 at $k = 1$ and 0.87–0.91 at $k = 200$ (`spectral_rotation_20261020`). The study read this as mixing within nearly tied groups, which is what Guided proof 5.11 and Exercise 5.14 predict for a power law. But the late head is flatter ($\alpha_{\text{top}} = 0.65$–$0.86$), so the relative gap below the top eigenvalue is $1 - 2^{-\alpha_{\text{top}}} \approx 0.36$–$0.45$, which is not small. Losing 16–22% of the top direction's curvature in 10 steps therefore means the 10-step change is not small compared with the top gap. Part II has to allow changes that are large at the top, not only mixing among near-ties. Rotation also exceeds batch noise: at fixed weights another batch's top-16 subspace captures 0.93–0.999.

6 Towards Part II

Part I treated one matrix. Part II lets it move. The first guided proof below is the equation of motion of spectrum and frame; you have already met it twice.

Guided proof 6.1 (How spectrum and frame move)**.** Let $A(t) = U(t)\Lambda(t)U(t)^\top$ be differentiable with distinct eigenvalues and differentiable $U(t) \in \mathrm{O}(D)$.

(a) Show that $W = U^\top U'$ is skew-symmetric.

(b) Show $U^\top A' U = \Lambda' + W\Lambda - \Lambda W$.

(c) Deduce $\lambda_k' = u_k^\top A' u_k$ and, for $i \ne j$, $W_{ij} = u_i^\top A' u_j / (\lambda_j - \lambda_i)$. Check that this is Guided proof 5.13 with $E = A'$.

(d) Interpret: "stable spectrum, rotating frame" means that A' is nearly off-diagonal in the current eigenbasis. The rotation rate of a pair of eigenvectors is the off-diagonal entry of A' divided by their gap.

Exercise 6.2 (Two fingerprints)**.** (a) (Isotropic noise.) Suppose that over a short time h the matrix changes by $\Delta A = \sigma\sqrt{h}\, G$, with G as in Exercise 5.14 and independent of the past. To first order, show that the rotation of the pair (i, j) over that time, hW_{ij}, has variance $\sigma^2 h/(\lambda_i - \lambda_j)^2$, while λ_k changes with variance $2\sigma^2 h$, independent of the gaps. Pairs with small gaps rotate fastest. (With the second-order terms this becomes Dyson Brownian motion; Tao 2012 and Allez & Bouchaud 2012.)

(b) (Frame diffusion.) Suppose instead $U' = UW$ with W a random skew-symmetric matrix whose law does not depend on Λ, and $\Lambda' = 0$. How large are the off-diagonal entries of $U^\top A' U$ now, as a function of the gaps?

(c) Which measurement on saved spectra and eigenvectors would tell the two pictures apart?

The plan for Part II.

(1) Three stochastic pictures with different fingerprints: isotropic matrix noise (rotation rate $\propto \text{gap}^{-2}$), frame diffusion (rate independent of the gaps) and deterministic drift (ballistic, hence predictable). Their fingerprints are capture against lag (linear or quadratic onset), rotation against gap, and replicas forked with different data orders.

(2) Structure: the change of a GGN under a weight step is not isotropic. What is its structure?

(3) Data: `spectral_rotation_20261020` saved spectra and captures; replays (10–12 minutes per trajectory) give eigenvectors. `noise_scale_20261021`'s forms give moments of the gradient's and the noise's spectral measures at 36 states without new compute.

What is known (searches of 23 October 2026)

References marked * were checked online on 23 October 2026, at least at the level of the abstract; the others are cited from memory.

- **EKFAC** (George et al. 2018*): the eigenbasis is recomputed rarely and the diagonal tracked every step, justified by cost and a Frobenius argument (Warm-up 5.20). No model of how the eigenbasis moves.
- **SOAP** (Vyas et al. 2024*): Adam in Shampoo's eigenbasis. The refresh interval is studied empirically; there is no model of eigenbasis drift against eigenvalue drift and no theory of the optimal interval.
- **Purifying Shampoo** (Eschenhagen et al. 2025*): decouples eigenvalue and eigenbasis updates and proposes an adaptive refresh criterion that controls the current approximation error. It does not forecast the error from a model of the frame's motion.
- **Power-law Hessian spectra**: Xie et al. (2022*) and Tang et al. (2025*). **Eigenvector dynamics**: Gur-Ari, Roberts & Dyer (2018); Marjankowska, Modugno & Barucca (2026*).
- **Random-matrix statistics**: rotationally invariant estimators (Ledoit & Péché 2011; Bun et al. 2016*; Bun, Bouchaud & Potters 2017*); eigenvector dynamics of exponentially weighted covariance matrices, the estimator family of SOAP's statistics (Allez & Bouchaud 2012).
- **Probabilistic numerics**: Hennig & Kiefel (2013); Hennig (2015); de Roos & Hennig (2019*); Wenger & Hennig (2020); Tatzel et al. (2025*).

As far as these searches show, no published work treats the spectral law as (nearly) stationary and the frame as a latent process, and derives SOAP's refresh schedule, averaging windows or treatment of momentum from such a model. Searches can miss work; this should be re-checked before any claim of novelty. The compact note, `spectral_frame.pdf`, Section 7, has the full list with comments.

Questions for us

Question 1. Which matrix should the theory treat first: the GGN, which we measured, or SOAP's own statistic, Kronecker factors of gradient second moments (Guided proof 2.12)?

Question 2. Is a scale-free law the right null hypothesis, given that the trace grows 1.3–1.6× during the learning-rate decay while the shape does not (`spectral_rotation_20261020`)?

Question 3. Which prior on the frame: uniform (Haar), or concentrated around SOAP's current basis (a matrix Bingham or von Mises–Fisher law)?

Question 4. Which first measurement: rotation against gap from replays (Exercise 6.2), or the gradient's spectral measure, which the forms saved by `noise_scale_20261021` give for free?

A Hints

Hints are short on purpose. If a hint gives too much away, read only its first sentence.

Section 1

Warm-up 1.3. (a) $\operatorname{tr}(XY) = \operatorname{tr}(YX)$, and $\|A\|_{\mathrm{F}}^2 = \operatorname{tr}(A^\top A)$. (b), (c) Write $v = Uz$; then $\|v\| = \|z\|$ and $v^\top A v = \sum_k \lambda_k z_k^2$. (d) For "only if", test $v = u_D$.

Warm-up 1.4. Compute $u^\top A w$ in two ways.

Exercise 1.5. (a) $A^i = U\Lambda^i U^\top$. (b) $p(A)$ is computed from A alone. (c) Powers of A commute with A. For the inverse, multiply $(A + \delta I)$ by $U \operatorname{diag}(1/(\lambda_k + \delta))U^\top$.

Warm-up 1.6. $(QAQ^\top)(Qu_k) = QAu_k$. For f, use the polynomial p of Exercise 1.5.

Guided proof 1.8. (a) Product rule at $t = 0$; $(e^{tW})^\top = e^{tW^\top}$. (b) Which entries of a skew-symmetric matrix are free? (c) Product rule, with $Q(0) = I$ and $Q'(0)^\top = -W$. (d) $(W\Lambda)_{ij} = W_{ij}\lambda_j$. (e) For the image: given S, try $W_{ij} = S_{ij}/(\lambda_j - \lambda_i)$. (f) $(\lambda_j - \lambda_i)W_{ij} = 0$ allows $W_{ij} \neq 0$ exactly when $\lambda_i = \lambda_j$. (g) $Q\Lambda = \Lambda Q$; compare entries as in (d).

Exercise 1.11. Choose $Q = U^\top$.

Warm-up 1.12. Test with $Q = U^\top$, where U is a frame of a 2×2 matrix with non-zero off-diagonal entries, or with a permutation matrix.

Guided proof 1.13. (a) $R_k^2 = I$. (b) Check $R_k A R_k$ on the basis $u_1, \ldots, u_D$. (c) Take $Q = R_k$ in the definition. (d) $R_k G(A) u_k = G(A) R_k u_k$. (e) Interpolate as in Exercise 1.5.

Exercise 1.14. $(QAQ^\top)^i = QA^iQ^\top$, so the numbers $b^\top A^i b$ do not change.

Section 2

Warm-up 2.2. N_A is a step function that jumps at $t = 3$, 1 and 0; decide which value it takes at each jump point.

Exercise 2.3. (a) The spectrum is sorted. (b) What is the set $\{t : N_A(t) \geq k\}$? (c) Divide by D.

Guided proof 2.6. (a) Sorted spectrum. (b) $t \geq \lambda_K = cK^{-\alpha}$. (c) Count integers. (d) Differentiate the tail and change its sign. (e) $t = ce^{-v}$. (f) Evaluate N_A at $t = ck^{-\alpha}$ and just above it.

Exercise 2.7. (c) $\lambda_k \geq \lambda_1/10$ if and only if $k \leq 10$; the trace share is H_{10}/H_{4096}.

Exercise 2.8. (b), (c) Functions of A keep its frame. (d) $P^{1/2} = \operatorname{diag}(\varepsilon^{-1/2}, 1)$.

Exercise 2.10. On the lower piece, solve $c\,k_b^{-\alpha_1}(k/k_b)^{-\alpha_2} \ge t$ for k.

Warm-up 2.11. Mixed-product rule $(X \otimes Y)(Z \otimes T) = (XZ) \otimes (YT)$, and $(v \otimes w)^\top (v' \otimes w') = (v^\top v')(w^\top w')$.

Guided proof 2.12. (a) For fixed i, which j are allowed? Why is $j \le n$ automatic? (b) $\sum_{i \le x} x/i = xH_{\lfloor x \rfloor}$. (c) Use $H_s = \ln s + \gamma + O(1/s)$ with $s = \lfloor \sqrt{x} \rfloor$, and $s^2 = x + O(\sqrt{x})$. (d) Take logarithms. (e) Compute $d \ln k / d \ln x$ for $k = x \ln x$.

Section 3

Warm-up 3.1. For $k \ge m$: $f(k+1) \le \int_k^{k+1} f \le f(k)$. Sum.

Guided proof 3.2. (a) Compare the tail with $\int_m^\infty x^{-s} dx$. (b) The warm-up with $f(x) = 1/x$ on one block. (c) The warm-up with $m = 1$; for the share, compute $S_s(K) - S_s(K/2)$. (d) For increasing f: $\int_{k-1}^k f \le f(k) \le \int_k^{k+1} f$.

Exercise 3.3. $\operatorname{tr} A^q$ is a power sum with $s = \alpha q$; Guided proof 3.2 has one case for each column. For $q = -1$, $s = -\alpha < 0$.

Exercise 3.4. (a) The eigenvalues in $[t, 2t)$ have ranks $c/(2t) < k \le c/t$. (b) Divide two harmonic numbers.

Warm-up 3.5. Split the sum at δ: $\lambda/(\lambda + \delta) \ge \frac{1}{2}$ if and only if $\lambda \ge \delta$, and $\lambda/(\lambda + \delta) \le \lambda/\delta$.

Guided proof 3.6. (a) $\lambda_k/(\lambda_k + \delta) = 1/(1 + \delta/\lambda_k)$. For the comparison, use $f(k) \le \int_{k-1}^k f$ and $\int_k^{k+1} f \le f(k)$, plus $\int_0^1 f \le 1$. (b) $dx = a^{-1/\alpha} du$. (c) $du = \frac{1}{\alpha} w^{1/\alpha - 1} dw$, so $s = 1/\alpha$; for $\alpha = 2$ the integral is an arctangent. (e) The same comparison on $[0, D]$.

Compute 3.7. 10^7 float64 values take 80 MB; summing them in one numpy call is fine.

Section 4

Warm-up 4.4. (a) A linear map of a Gaussian vector is Gaussian; compute the covariance. (b) $R(Q^\top v) = (QR^\top)^\top v$. The uniform law is the only rotation-invariant law on the sphere. (c) Take R a permutation matrix in (b).

Guided proof 4.5. (a) $f(A) = Q f(\Lambda) Q^\top$. (b) $g_k = \|g\| \psi_k$. (c) Expand $\|g\|^4 = (\sum_k g_k^2)^2$. (d) $\operatorname{Var} X = \mathbb{E}X^2 - (\mathbb{E}X)^2$. (e) $\operatorname{Var} \sum_k f_k w_k = \sum_{k,l} f_k f_l \operatorname{Cov}(w_k, w_l)$; separate $k = l$ from $k \ne l$. (f) Use $\sum_k k^{-2} \approx \zeta(2)$, or compute numerically.

Exercise 4.6. (a) Change variables $y = g^2$ in the Gaussian density. (c) $a_k = \frac{1}{2}$ and $a_0 = D/2$.

Exercise 4.7. Average $\Pr(I = k \mid Q) = w_k$ over Q.

Compute 4.8. Draw g in batches of, say, 20,000 rows to keep memory small. Use the population variance of the eigenvalues for $\operatorname{Var}_{\mu_A}$.

Warm-up 4.11. (a) $f(A) = \sum_k f(\lambda_k) u_k u_k^\top$. (b) The eigenvectors of $QAQ^\top$ are Qu_k. (c) Warm-up 4.4(c). (d) Here $u_k = e_k$.

Guided proof 4.14. (a) Induction: multiplying by a tridiagonal matrix extends the support by one position. (b) $AV_m T_m^j e_1 = V_m T_m^{j+1} e_1 + \beta_m v_{m+1} (e_m^\top T_m^j e_1)$; when is the last factor zero? (c) $v_1^\top A^{2j} v_1 = \|A^j v_1\|^2$ and $v_1^\top A^{2j+1} v_1 = (A^j v_1)^\top A (A^j v_1)$, with $V_m^\top A V_m = T_m$. (d) $e_1^\top T_m^i e_1 = e_1^\top S \Theta^i S^\top e_1$.

Guided proof 4.15. (a) $b = Ax$. (b) Functions of A commute. (d) Choose $p(\lambda) = \prod_j (1 - \lambda/\lambda_j)$ over the atoms.

Exercise 4.16. The objective is a quadratic in η; then use Warm-up 4.11(a).

Guided proof 4.17. (a) Insert $f = 1/\lambda$ and $f = 1$ into Exercise 4.16 and divide. (b) $1 = (\mathbb{E}[\lambda^{1/2}\lambda^{-1/2}])^2$. (c) Expand and divide by $\lambda > 0$. (d) Apply $2\sqrt{xy} \le x + y$ to $x = \mathbb{E}_\nu \lambda$ and $y = mM\,\mathbb{E}_\nu \lambda^{-1}$.

Exercise 4.18. $\mathbb{E}_\mu \lambda = \frac{1}{D}\sum_k c/k$ and $\mathbb{E}_\mu \lambda^{-1} = \frac{1}{D}\sum_k k/c$.

Section 5

Warm-up 5.1. (a) Characteristic polynomial. (b) Write the two rows of the eigenvector equation, multiply them by $\sin\theta$ and $\cos\theta$, and subtract. (d) $\tan 2\theta \to \infty$.

Guided proof 5.2. (a) $v = \sum_{i \le k} z_i u_i$. (b) $k + (D - k + 1) > D$. (c) $v = \sum_{i \ge k} z_i u_i$.

Guided proof 5.3. (a) $v^\top E v \le \|E\|_{\mathrm{op}}$ for unit v. (b) Apply (a) to every v in every S.

Exercise 5.5. Swapping an inversion changes the sum by $(a_i - a_j)(b_{\pi(j)} - b_{\pi(i)})$.

Guided proof 5.6. (b) $\mathrm{tr}(AB) = \mathrm{tr}(\mathrm{diag}(a)\, W\, \mathrm{diag}(b)\, W^\top)$. (c) Rows and columns of an orthogonal matrix are unit vectors. (d) $\mathrm{tr}(AB)$ is linear in P.

Exercise 5.7. For a permutation coupling the cost is $\frac{1}{D}\sum_i (a_i - b_{\pi(i)})^2$; expand the square.

Guided proof 5.8. (b) $Auu^\top = \lambda_k uu^\top = uu^\top A$. (c) Expand z in the eigenvectors u_j, $j \ne k$. (d) $\|(A - \mu I)Pv\| = \|PEv\|$. (e) Consider $\|E\|_{\mathrm{op}} \le g_k/2$ and $\|E\|_{\mathrm{op}} > g_k/2$ separately.

Exercise 5.10. (a) Concave functions lie below their tangents. (b) $(\lambda_k - \lambda_{k+1})/\lambda_k = h(1/k)$. (c) $x \mapsto cx^{-\alpha}$ is convex.

Guided proof 5.11. (a) Weyl. (b) $(1 + 1/k)^{-\alpha-1} \ge 2^{-\alpha-1}$. (c) Solve $2\eta/g_k \le s$ for k. (d) The ratio is a power of η.

Exercise 5.12. With $\alpha = c = 1$: $K_{\mathrm{val}} = s/\eta$ and $K_{\mathrm{vec}} = (s/(8\eta))^{1/2}$. On which pair of directions would a worst-case E put its size?

Guided proof 5.13. (b) $u_k^\top A = \lambda_k u_k^\top$, so two terms cancel. (c) $u_j^\top u_k = 0$. (d) Expand u_k' in the basis $u_1, \ldots, u_D$.

Exercise 5.14. (a) Take $Q = U$. (b) For $|m| \ll k$, $\lambda_k - \lambda_{k+m} \approx -m g_k$, and $\sum_{m \ne 0} m^{-2} = 2\zeta(2)$; for $j \gg k$, $(\lambda_k - \lambda_j)^{-2} \approx \lambda_k^{-2}$. (d) Set $\varepsilon\sqrt{2\zeta(2)}k^{\alpha+1}/(\alpha c) = s$.

Compute 5.15. $(Z + Z^\top)/\sqrt{2}$ has off-diagonal variance 1 and diagonal variance 2. The function `eigh` in `numpy.linalg` returns ascending eigenvalues; reverse them. The sine of the k-th angle is $\sqrt{1 - (e_k^\top v_k)^2}$.

Guided proof 5.16. (a) $U_A^\top A = \mathrm{diag}(a) U_A^\top$ and $BU_B = U_B\,\mathrm{diag}(b)$. (c) Entrywise, $|f(a_i) - f(b_j)| \le L|a_i - b_j|$.

Exercise 5.17. Bound the derivative.

Exercise 5.18. $\|P - P'\|_{\mathrm{F}}^2 = \mathrm{tr}\, P + \mathrm{tr}\, P' - 2\,\mathrm{tr}(PP')$ and $\mathrm{tr}(PP') = \|V^\top V'\|_{\mathrm{F}}^2$.

Warm-up 5.20. $\|A - UMU^\top\|_{\mathrm{F}} = \|U^\top A U - M\|_{\mathrm{F}}$.

Guided proof 5.21. (a) Expand x_i in the eigenvectors. (b) Bessel's inequality. (c) Moving weight to a larger λ_j never decreases the sum. (e) $k = 1$, and the case $k = D - 1$ subtracted from the trace.

Section 6

Guided proof 6.1. (a) Differentiate $U^\top U = I$. (b) Differentiate $A = U\Lambda U^\top$ and multiply by $U^\top$ on the left and U on the right. (c) Guided proof 1.81.8(d).

Exercise 6.2. (a) Exercise 5.14(a) with ΔA in place of E. (b) Read off $(U^\top A' U)_{ij} = (\lambda_j - \lambda_i)W_{ij}$. (c) Plot a measured rotation against the gap.

B Solutions

Section 1

Warm-up 1.3. (a) $\operatorname{tr} A = \operatorname{tr}(U\Lambda U^\top) = \operatorname{tr}(\Lambda U^\top U) = \operatorname{tr}\Lambda$. Likewise $\|A\|_{\mathrm{F}}^2 = \operatorname{tr}(A^\top A)$, which equals $\operatorname{tr}(U\Lambda^2 U^\top) = \sum_k \lambda_k^2$.
(b) For $v = Uz$,

$$\|Av\| = \|U\Lambda z\| = \|\Lambda z\| \le \max_k |\lambda_k|\,\|z\| = \max_k |\lambda_k|\,\|v\|,$$

with equality for $z = e_k$ at the largest $|\lambda_k|$.
(c) $v^\top Av = z^\top \Lambda z = \sum_k \lambda_k z_k^2$ lies between $\lambda_D \sum_k z_k^2$ and $\lambda_1 \sum_k z_k^2$, and $\sum_k z_k^2 = \|v\|^2$; equality at $z = e_D$ and $z = e_1$.
(d) If $\lambda_D \ge 0$, (c) gives $v^\top Av \ge 0$. If A is PSD, $\lambda_D = u_D^\top A u_D \ge 0$.

Warm-up 1.4. $u^\top Aw = \mu\, u^\top w$ and, by symmetry, $u^\top Aw = (Au)^\top w = \lambda\, u^\top w$. So $(\lambda - \mu)u^\top w = 0$.

Exercise 1.5. (a) $A^i = U\Lambda^i U^\top$ by induction, so $p(A) = Up(\Lambda)U^\top = U\operatorname{diag}(p(\lambda_k))U^\top = U\operatorname{diag}(f(\lambda_k))U^\top$.
(b) p depends only on f and the values λ_k, and $p(A)$ is computed from A without any frame.
(c) $p(A)A = Ap(A)$ because powers of A commute. A^2 is $f(A)$ for $f(\lambda) = \lambda^2$. For $\delta > -\lambda_D$, $(A + \delta I)\,U\operatorname{diag}(1/(\lambda_k + \delta))U^\top = U\operatorname{diag}((\lambda_k + \delta)/(\lambda_k + \delta))U^\top = I$, so the inverse is $f(A)$ with $f(\lambda) = 1/(\lambda + \delta)$. For PSD A, $U\operatorname{diag}(\lambda_k^{1/2})U^\top$ is PSD and squares to A; $(A + \delta I)^{-\nu}$ is defined as $f(A)$ with $f(\lambda) = (\lambda + \delta)^{-\nu}$, which agrees with the inverse for $\nu = 1$.

Warm-up 1.6. $\|Qv\|^2 = v^\top Q^\top Qv = \|v\|^2$. $(QAQ^\top)(Qu_k) = QAu_k = \lambda_k Qu_k$, and the Qu_k are orthonormal, so $QAQ^\top = (QU)\Lambda(QU)^\top$ has spectrum $\sigma(A)$ and frame QU. Hence $f(QAQ^\top) = QUf(\Lambda)U^\top Q^\top = Qf(A)Q^\top$.

Guided proof 1.8. (a) $\frac{d}{dt}(Q^\top Q) = Q'^\top Q + Q^\top Q' = 0$; at $t = 0$ this is $W^\top + W = 0$. For skew W, $(e^{tW})^\top = e^{tW^\top} = e^{-tW}$, and $e^{-tW}e^{tW} = I$ because tW and $-tW$ commute.
(b) $W_{ii} = -W_{ii}$ forces a zero diagonal, and the entries below the diagonal are fixed by those above, which are free: $D(D-1)/2$ numbers.
(c) $A' = Q'\Lambda Q^\top + Q\Lambda Q'^\top$, which at $t = 0$ is $W\Lambda + \Lambda W^\top = W\Lambda - \Lambda W$.
(d) $(W\Lambda)_{ij} = W_{ij}\lambda_j$ and $(\Lambda W)_{ij} = \lambda_i W_{ij}$.
(e) If $(\lambda_j - \lambda_i)W_{ij} = 0$ for all $i \ne j$ and the λ's are distinct, every off-diagonal W_{ij} vanishes; the diagonal vanishes anyway, so $W = 0$. The image is symmetric, since $(W\Lambda - \Lambda W)^\top = -\Lambda W + W\Lambda$, and has zero diagonal by (d). Conversely, for symmetric S with zero diagonal, $W_{ij} = S_{ij}/(\lambda_j - \lambda_i)$ $(i \ne j)$ is skew-symmetric and maps to S. So the orbit has $D(D-1)/2$ independent directions at Λ, the diagonal matrices add D more, and together they span $\mathbb{S}^D$, of dimension $D(D+1)/2$.
(f) $W_{ij}(\lambda_j - \lambda_i) = 0$ for all i, j allows $W_{ij} \ne 0$ exactly when $\lambda_i = \lambda_j$, that is, inside the diagonal blocks of equal values. Inside a block of size m_j, W is an arbitrary skew-symmetric $m_j \times m_j$ matrix, so the kernel has dimension $\sum_j m_j(m_j-1)/2$ and the image $D(D-1)/2 - \sum_j m_j(m_j-1)/2$.
(g) $Q\Lambda Q^\top = \Lambda$ means $Q\Lambda = \Lambda Q$, that is, $Q_{ij}(\lambda_j - \lambda_i) = 0$, so Q is diagonal; a diagonal orthogonal matrix satisfies $Q_{ii}^2 = 1$. If $U\Lambda U^\top = V\Lambda V^\top$, then $Q = U^\top V$ satisfies $Q\Lambda Q^\top = \Lambda$, so $V = UQ$: the columns agree up to sign.

Exercise 1.11. If F is invariant, $F(A) = F(U^\top AU) = F(\operatorname{diag}\sigma(A))$. Conversely, if $F(A) = \varphi(\sigma(A))$, then $F(QAQ^\top) = \varphi(\sigma(QAQ^\top)) = \varphi(\sigma(A))$ by Warm-up 1.6.

Warm-up 1.12. (a) $\operatorname{tr} A$, $\det A$, $\lambda_1(A)$ and $\|A\|_{\mathrm{F}}$ are functions of the spectrum, hence invariant. A_{11} and $v^\top Av$ are not: for $A = \operatorname{diag}(1,0)$ and Q the swap of the two coordinates, $(QAQ^\top)_{11} = 0 \neq 1$.
(b) A^2, $(A+\delta I)^{-1}$ and $\Pi_k(A)$ are equivariant, by Warm-up 1.6: each is a function of A (for Π_k, the indicator of $[\lambda_k, \infty)$). $\operatorname{Diag}(A)$ is not: for $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $\operatorname{Diag}(A) = 0$, but with $Q = U^\top$ (U a frame of A), $QAQ^\top = \operatorname{diag}(1,-1)$ has $\operatorname{Diag}(QAQ^\top) = \operatorname{diag}(1,-1) \neq Q\,0\,Q^\top$. $A + B_0$ is equivariant only if $QB_0Q^\top = B_0$ for all Q, which forces $B_0 = cI$.

Guided proof 1.13. (a) $R_k^\top = R_k$ and $R_k^2 = I - 4u_ku_k^\top + 4u_k(u_k^\top u_k)u_k^\top = I$. $R_ku_k = u_k - 2u_k = -u_k$ and $R_ku_j = u_j$ for $j \neq k$. Since the u_j form a basis, the -1 eigenspace is $\operatorname{span}(u_k)$.
(b) $R_kAR_ku_j = R_kA(\pm u_j) = \pm\lambda_jR_ku_j = \lambda_ju_j = Au_j$ for every j.
(c) Equivariance with $Q = R_k$: $G(A) = G(R_kAR_k^\top) = R_kG(A)R_k$.
(d) By (c), $G(A)u_k = R_kG(A)R_ku_k = -R_kG(A)u_k$, so $R_k(G(A)u_k) = -G(A)u_k$ and $G(A)u_k \in \operatorname{span}(u_k)$.
(e) $G(A)U = U\operatorname{diag}(\varphi)$, so $G(A) = U\operatorname{diag}(\varphi)U^\top$, which commutes with $A = U\Lambda U^\top$. For distinct eigenvalues choose p with $p(\lambda_k) = \varphi_k$; then $p(A) = U\operatorname{diag}(\varphi)U^\top$.

Exercise 1.14. $(Qb)^\top(QAQ^\top)^i(Qb) = b^\top Q^\top QA^iQ^\top Qb = b^\top A^ib$, so $p_{QAQ^\top,Qb} = p_{A,b} =: p$ and $F(QAQ^\top, Qb) = p(QAQ^\top)Qb = Qp(A)Q^\top Qb = QF(A,b)$. With a stored frame, $x' = U_{\text{old}}\operatorname{diag}(\varphi)U_{\text{old}}^\top Qb$ is in general not Qx: the result depends on how A and b sit relative to U_{old}. The stored frame is a third argument, and only rotating it along (QU_{old}) restores equivariance.

Section 2

Warm-up 2.2. $\mu_A = \frac{1}{4}\delta_3 + \frac{1}{2}\delta_1 + \frac{1}{4}\delta_0$. $N_A(t) = 4$ for $t \leq 0$, 3 for $0 < t \leq 1$, 1 for $1 < t \leq 3$ and 0 for $t > 3$; it jumps down just after each eigenvalue. The rank function is $(3,1,1,0)$. At $t = 1$, $N_A = 3$: for $k = 2, 3$ both $\lambda_k = 1 \geq 1$ and $3 \geq k$ hold; for $k = 4$ both fail.

Exercise 2.3. (a) If $\lambda_k \geq t$, then $\lambda_1, \ldots, \lambda_k \geq t$, so $N_A(t) \geq k$. If at least k eigenvalues are $\geq t$, the k largest are, and in particular λ_k.
(b) By (a), $\{t : N_A(t) \geq k\} = (-\infty, \lambda_k]$, whose maximum is λ_k.
(c) $\mu_A([t,\infty)) \geq k/D$ if and only if $N_A(t) \geq k$, so $Q_A(k/D) = \max\{t : N_A(t) \geq k\} = \lambda_k$.

Guided proof 2.6. (a) For $k > K$: $\lambda_k \leq \lambda_{K+1} < \lambda_K \leq t$.
(b) $ck^{-\alpha} \geq t \iff k^\alpha \leq c/t \iff k \leq (c/t)^{1/\alpha}$, and $t \geq \lambda_K = cK^{-\alpha}$ gives $(c/t)^{1/\alpha} \leq K$.
(c) $N_A(t)$ counts the integers $1 \leq k \leq (c/t)^{1/\alpha}$, all of which are $\leq K$.
(d) Divide by D. Without the floor, $\mu_A([t,\infty)) = (c/t)^{1/\alpha}/D = \frac{K}{D}(t/\lambda_K)^{-1/\alpha}$ on $[\lambda_K, c]$: K/D times the tail of a Pareto law with index $1/\alpha$ and minimum λ_K. The density is $-\frac{d}{dt}[(c/t)^{1/\alpha}/D] = \frac{1}{\alpha}c^{1/\alpha}t^{-1/\alpha-1}/D$.
(e) With $t = ce^{-v}$, $(c/t)^{1/\alpha} = e^{v/\alpha}$, and $t \geq \lambda_K \iff v \leq \alpha \ln K$.
(f) Let $k \leq K$ and $t_k = ck^{-\alpha} \geq cK^{-\alpha}$. Then $N_A(t_k) = k$, so $\lambda_k \geq t_k$ by Exercise 2.32.3(a). For $t > t_k$, $(c/t)^{1/\alpha} < k$, so $N_A(t) < k$ and $\lambda_k < t$. Hence $\lambda_k = t_k$.

Exercise 2.7. (a) $c(2k)^{-\alpha}/(ck^{-\alpha}) = 2^{-\alpha}$.
(b) $(c/(t/2))^{1/\alpha} = 2^{1/\alpha}(c/t)^{1/\alpha}$.
(c) $\lambda_k = c/k \geq c/10$ if and only if $k \leq 10$: $10/4096 = 0.24\%$ of the directions, carrying $H_{10}/H_{4096} = 0.329$ of the trace. Within a factor 1,000: $k \leq 1000$, that is 24% of the directions, carrying $H_{1000}/H_{4096} = 0.842$. The bottom three quarters of the directions carry the remaining sixth of the trace.

Exercise 2.8. (a) $s\lambda_k = (sc)k^{-\alpha}$.
(b) A^p has eigenvalues $\lambda_k^p = c^p k^{-\alpha p}$, in the same frame and the same order.
(c) $P^{1/2} = A^{-\nu/2}$, and all three matrices are functions of A with the frame U, so $P^{1/2}AP^{1/2} = U\Lambda^{-\nu/2}\Lambda\Lambda^{-\nu/2}U^\top = A^{1-\nu}$. Exponent $\alpha(1-\nu)$, scale $c^{1-\nu}$, condition number $K^{\alpha(1-\nu)}$ on the window.
(d) $P^{1/2} = \mathrm{diag}(\varepsilon^{-1/2}, 1)$ and $P^{1/2}AP^{1/2} = \mathrm{diag}(1/\varepsilon, \varepsilon)$, with condition number ε^{-2} against ε^{-1} for A: the wrong frame squares the condition number instead of reducing it. Statements like (c) use that P and A share a frame. Otherwise the result depends on the relative frame, not on the two spectra alone.

Exercise 2.10. For $t \geq \lambda_{k_b} = ck_b^{-\alpha_1}$, $N_A(t) = (c/t)^{1/\alpha_1}$ as in Guided proof 2.6. For $\lambda_K \leq t < \lambda_{k_b}$, $\lambda_{k_b}(k/k_b)^{-\alpha_2} \geq t \iff k \leq k_b(\lambda_{k_b}/t)^{1/\alpha_2}$, so $N_A(t) = k_b(\lambda_{k_b}/t)^{1/\alpha_2}$. Both formulas give k_b at $t = \lambda_{k_b}$. So μ_A has a Pareto piece with tail index $1/\alpha_1$ on $[\lambda_{k_b}, c]$ and one with tail index $1/\alpha_2$ on $[\lambda_K, \lambda_{k_b})$.

Warm-up 2.11. The mixed-product rule gives $(R \otimes L)(v \otimes w) = (Rv) \otimes (Lw)$, and for eigenvectors this is $(r_j v_j) \otimes (\ell_i w_i) = r_j \ell_i (v_j \otimes w_i)$. Also $(v_j \otimes w_i)^\top (v_{j'} \otimes w_{i'}) = (v_j^\top v_{j'})(w_i^\top w_{i'})$, which is 1 if $(j,i) = (j',i')$ and 0 otherwise: mn orthonormal vectors in $\mathbb{R}^{mn}$, a full frame.

Guided proof 2.12. (a) $(ij)^{-\alpha} \geq t \iff ij \leq x$. For each $i \leq x$ the allowed j are $1, \ldots, \lfloor x/i \rfloor$, all $\leq n$ because $x \leq n$; and $i \leq x \leq m$.
(b) $\sum_{i \leq x}(x/i - 1) < N \leq \sum_{i \leq x} x/i$, that is, $xH_{\lfloor x \rfloor} - \lfloor x \rfloor < N \leq xH_{\lfloor x \rfloor}$, and $H_{\lfloor x \rfloor} = \ln x + O(1)$.
(c) If $i > \sqrt{x}$ and $j > \sqrt{x}$, then $ij > x$; so every counted point has $i \leq s$ or $j \leq s$, with $s = \lfloor \sqrt{x} \rfloor$. The strip $i \leq s$ holds $\sum_{i \leq s} \lfloor x/i \rfloor$ points, the strip $j \leq s$ as many, and both strips share the s^2 points with $i, j \leq s$ (all counted, since $ij \leq s^2 \leq x$). Then $\sum_{i \leq s} \lfloor x/i \rfloor = xH_s + O(s) = x(\frac{1}{2}\ln x + \gamma) + O(\sqrt{x})$, using $\ln s = \frac{1}{2}\ln x + O(1/\sqrt{x})$, and $s^2 = x + O(\sqrt{x})$. So $N = x \ln x + 2\gamma x - x + O(\sqrt{x})$.
(d) $x \ln x = k$ gives $x = k/\ln x$, and taking logarithms $\ln x = \ln k - \ln\ln x = \ln k\,(1 + o(1))$. So $x = (k/\ln k)(1 + o(1))$ and $\lambda_k = x^{-\alpha} \approx (k/\ln k)^{-\alpha}$.
(e) $\ln k = \ln x + \ln\ln x$, so $d\ln k / d\ln x = 1 + 1/\ln x$; with $\ln\lambda = -\alpha \ln x$, $d\ln\lambda / d\ln k = -\alpha/(1 + 1/\ln x)$.

Compute 2.13.

```
lam = np.sort(np.outer(1/np.arange(1, 65), 1/np.arange(1, 257)).ravel())[::-1]
k = np.arange(17, 201)
slope = np.polyfit(np.log(k), np.log(lam[k-1]), 1)[0]          # about -0.77
km = np.exp(np.mean(np.log(k))); x = km/np.log(km)
predicted = 1/(1+1/np.log(x))                                  # about 0.75
```

Section 3

Warm-up 3.1. For $x \in [k, k+1]$, $f(k+1) \leq f(x) \leq f(k)$; integrate: $f(k+1) \leq \int_k^{k+1} f \leq f(k)$. Summing the right inequality over $k = m, \ldots, M$ gives the lower bound. Summing the left one over $k = m, \ldots, M-1$ gives $\sum_{k=m+1}^{M} f(k) \leq \int_m^M f$; add $f(m)$. With $f(x) = 1/x$, $m = 1$ and $M = n$: $\ln(n+1) \leq H_n \leq 1 + \ln n$.

Guided proof 3.2. (a) $\sum_{k>m} k^{-s} \leq \int_m^\infty x^{-s}dx = m^{1-s}/(s-1)$; divide by $\zeta(s)$. For $s = 2$, $m = 16$: $(1/16)/\zeta(2) = 0.038$, against the true 0.037.
(b) With $f(x) = 1/x$ on the block, $\ln 2 = \int_{2^j}^{2^{j+1}} dx/x \leq \sum_{k=2^j}^{2^{j+1}-1} 1/k \leq 2^{-j} + \int_{2^j}^{2^{j+1}-1} dx/x \leq 2^{-j} + \ln 2$.

(c) With $m = 1$, $M = K$: $((K+1)^{1-s} - 1)/(1-s) \le S_s(K) \le 1 + (K^{1-s} - 1)/(1-s)$, and $0 \le (K+1)^{1-s} - K^{1-s} \le (1-s)K^{-s} \le 1 - s$ by the mean value theorem; so both bounds are $K^{1-s}/(1-s) + O(1)$. Then $S_s(K) - S_s(K/2) = (K^{1-s} - (K/2)^{1-s})/(1-s) + O(1)$, and dividing by $S_s(K)$, with $K^{1-s} \to \infty$, gives $1 - 2^{s-1} + o(1)$.
(d) For increasing $f(x) = x^{-s} = x^{|s|}$: $\int_{k-1}^{k} f \le f(k) \le \int_{k}^{k+1} f$, so $K^{1-s}/(1-s) = \int_0^K f \le S_s(K) \le \int_1^{K+1} f \le (K+1)^{1-s}/(1-s)$, and both bounds are $K^{1-s}/(1-s)\,(1+o(1))$.

Exercise 3.3. With $s = \alpha q$: head-dominated if $\alpha q > 1$ (case (a)); marginal at $\alpha q = 1$, with equal contributions per octave (case (b)); tail-dominated if $\alpha q < 1$, where the ranks above $D/2$ keep a fixed share (cases (c) and (d)).

Functional	head when	marginal at	tail when
$\operatorname{tr} A^2$	$\alpha > 1/2$	$\alpha = 1/2$	$\alpha < 1/2$
$\operatorname{tr} A$	$\alpha > 1$	$\alpha = 1$	$\alpha < 1$
$\operatorname{tr} A^{1/2}$	$\alpha > 2$	$\alpha = 2$	$\alpha < 2$
$\operatorname{tr} A^{-1}$	never	—	always; $\approx c^{-1} D^{1+\alpha}/(1+\alpha)$

At $\alpha \approx 1$ a fit on ranks 1–200 decides $\operatorname{tr} A^2$ (by (a), ranks beyond 200 carry at most about 0.3% of it), decides $\operatorname{tr} A$ only up to the logarithm of the unseen ranks, and decides neither $\operatorname{tr} A^{1/2}$ nor any negative power.

Exercise 3.4. (a) $c/k \in [t, 2t) \iff c/(2t) < k \le c/t$: one dyadic block of ranks, up to rounding at its ends, starting near $a = c/(2t)$. By Guided proof 3.2(b) its sum is $c(\ln 2 + O(1/a)) = c \ln 2 + O(t)$.
(b) $\sum_{j \le k}(c/j) / \sum_{j \le D}(c/j) = H_k / H_D$, and $H_n \approx \ln n$.
(c) $H_{16} = 3.381$, $H_{640} = 7.039$, $H_{4096} = 8.895$ and $H_{16384} = 10.281$ give 0.480, 0.380 and 0.329.

Warm-up 3.5. Terms with $\lambda_k \ge \delta$ lie in $[\frac{1}{2}, 1]$, and there are $N_A(\delta)$ of them. Terms with $\lambda_k < \delta$ lie in $[0, \lambda_k/\delta]$. Add.

Guided proof 3.6. (a) $\lambda_k/(\lambda_k + \delta) = 1/(1 + \delta/\lambda_k) = 1/(1 + ak^\alpha) = f(k)$. f decreases on $[0, \infty)$ and $f(0) = 1$. For $k \ge 1$, $f(k) \le \int_{k-1}^{k} f$, so $\sum_{k \ge 1} f(k) \le \int_0^\infty f$; and $\int_k^{k+1} f \le f(k)$ with $\int_0^1 f \le 1$ gives $\int_0^\infty f \le 1 + \sum_{k \ge 1} f(k)$. The integral is finite because $f(x) \sim (ax^\alpha)^{-1}$ and $\alpha > 1$.
(b) $x = a^{-1/\alpha} u$ turns ax^α into u^α and dx into $a^{-1/\alpha} du$.
(c) $u = w^{1/\alpha}$, $du = \frac{1}{\alpha} w^{1/\alpha - 1} dw$, so $\int_0^\infty du/(1+u^\alpha) = \frac{1}{\alpha} \int_0^\infty w^{1/\alpha - 1}/(1+w)\, dw = (\pi/\alpha)/\sin(\pi/\alpha)$ with $s = 1/\alpha \in (0, 1)$. For $\alpha = 2$: $\int_0^\infty du/(1+u^2) = \pi/2 = (\pi/2)/\sin(\pi/2)$.
(d) $a^{-1/\alpha} = (c/\delta)^{1/\alpha}$; combine with (a). The factor exceeds 1 because $\sin y < y$; it is $\pi/2 \approx 1.57$ at $\alpha = 2$, tends to 1 as $\alpha \to \infty$ and to ∞ as $\alpha \downarrow 1$.
(e) With $f(x) = 1/(1 + \delta x/c)$: $\sum_{k=1}^{D} f(k) \le \int_0^D f \le 1 + \sum_{k=1}^{D-1} f(k)$, and $\int_0^D dx/(1 + \delta x/c) = (c/\delta) \ln(1 + D\delta/c)$.

Compute 3.7.

```
k = np.arange(1, 10**7+1, dtype=float)
print(np.sum(1/(1+1e-6*k**2)), 1e3*np.pi/2)                    # alpha = 2
k = np.arange(1, 4097, dtype=float)
print(np.sum(1/(1+1e-3*k)), 1e3*np.log(1+4.096))               # alpha = 1
```

Section 4

Warm-up 4.4. (a) Rg is Gaussian with mean 0 and covariance $RIR^\top = I$.
(b) For fixed R, $R(Q^\top v) = (QR^\top)^\top v$, and $QR^\top$ is Haar distributed. So $Q^\top v$ and $RQ^\top v$ have the same law for every R: the law of $Q^\top v$ on the sphere is rotation invariant, hence uniform.
(c) $\sum_k w_k = \|Q^\top v\|^2 = 1$. With R a permutation matrix, (b) shows that the coordinates of $Q^\top v$ are exchangeable, so all $\mathbb{E}w_k$ are equal, and they sum to 1.

Guided proof 4.5. (a) $v^\top f(A)v = v^\top Q f(\Lambda)Q^\top v = \psi^\top f(\Lambda)\psi = \sum_k f(\lambda_k)\psi_k^2$.
(b) $g_k = \|g\|\psi_k$, so $g_k^4 = \|g\|^4 w_k^2$ and $g_k^2 g_l^2 = \|g\|^4 w_k w_l$; since ψ and $\|g\|$ are independent, the expectations factor.
(c) $\mathbb{E}\|g\|^4 = \sum_k \mathbb{E}g_k^4 + \sum_{k\neq l} \mathbb{E}g_k^2\,\mathbb{E}g_l^2 = 3D + D(D-1) = D(D+2)$. Divide.
(d) $\operatorname{Var} w_k = \frac{3}{D(D+2)} - \frac{1}{D^2} = \frac{2(D-1)}{D^2(D+2)}$ and $\operatorname{Cov}(w_k, w_l) = \frac{1}{D(D+2)} - \frac{1}{D^2} = -\frac{2}{D^2(D+2)}$.
(e) $\mathbb{E}\sum_k f_k w_k = \frac{1}{D}\sum_k f_k = \int f\, d\mu_A$. With $f_k = f(\lambda_k)$,

$$\begin{aligned}\operatorname{Var}\sum_k f_k w_k &= \frac{2}{D^2(D+2)}\Big[(D-1)\sum_k f_k^2 - \sum_{k\neq l} f_k f_l\Big] \\ &= \frac{2}{D^2(D+2)}\Big[D\sum_k f_k^2 - \Big(\sum_k f_k\Big)^2\Big] = \frac{2\operatorname{Var}_{\mu_A}(f)}{D+2}.\end{aligned}$$

(f) For $\lambda_k = 1/k$ and $D = 4{,}096$: mean $H_D/D = 2.17 \times 10^{-3}$, second moment $\frac{1}{D}\sum_k k^{-2} \approx \zeta(2)/D = 4.0 \times 10^{-4}$, standard deviation 0.0199, coefficient of variation 9.2; times $\sqrt{2/4098}$ this is 0.203. The spectrum spans a factor of 4,096, yet the Rayleigh quotient of a random direction is within about 20% of $\operatorname{tr} A/D$.

Exercise 4.6. (a) Differentiate $\Pr(g^2 \le y) = \Pr(|g| \le \sqrt{y})$ in y. The density of g^2 is

$$\frac{y^{-1/2}e^{-y/2}}{\sqrt{2\pi}} = \frac{y^{a-1}e^{-y/\theta}}{\Gamma(a)\,\theta^a} \quad \text{with } a = \tfrac{1}{2},\ \theta = 2,$$

because $\Gamma(\frac{1}{2}) = \sqrt{\pi}$: the Gamma density.
(b) $w_k = g_k^2/\sum_l g_l^2$ with independent Gamma$(\frac{1}{2}, 2)$ numerators.
(c) With $a_k = \frac{1}{2}$ and $a_0 = D/2$: $\mathbb{E}w_k = 1/D$, and

$$\operatorname{Var} w_k = \frac{\frac{1}{2}(\frac{D}{2} - \frac{1}{2})}{\frac{D^2}{4}(\frac{D}{2}+1)} = \frac{2(D-1)}{D^2(D+2)}, \qquad \operatorname{Cov}(w_k, w_l) = -\frac{\frac{1}{4}}{\frac{D^2}{4}(\frac{D}{2}+1)} = -\frac{2}{D^2(D+2)},$$

as in Guided proof 4.5(d).

Exercise 4.7. $\Pr(I = k) = \mathbb{E}[\Pr(I = k \mid Q)] = \mathbb{E}w_k = 1/D$, so $\lambda_I \sim \mu_A$. And $\mathbb{E}[f(\lambda_I) \mid Q] = \sum_k w_k f(\lambda_k) = v^\top f(A)v$ by Guided proof 4.5(a).

Compute 4.8.

```
rng = np.random.default_rng(0); D = 256; lam = 1/np.arange(1, D+1)
vals = []
for _ in range(10):
    g = rng.standard_normal((20_000, D))
    vals.append((g**2 @ lam)/np.sum(g**2, 1))
vals = np.concatenate(vals)
print(vals.var()/(2*lam.var()/(D+2)))   # close to 1
print(vals.mean()/lam.mean())            # close to 1
```

Warm-up 4.11. (a) Since $f(A) = \sum_k f(\lambda_k) u_k u_k^\top$, we get $v^\top f(A) v = \sum_k f(\lambda_k)(u_k^\top v)^2 = \int f \, d\mu_{A,v}$. With $f = 1$ the mass is $\sum_k (u_k^\top v)^2 = \|v\|^2$.
(b) $QAQ^\top$ has eigenvalues λ_k with eigenvectors Qu_k, and $(Qu_k)^\top Qv = u_k^\top v$.
(c) In the Haar model $u_k = q_k$, and $\mathbb{E}(q_k^\top v)^2 = \|v\|^2/D$ by Warm-up 4.4(c) applied to $v/\|v\|$. So every atom λ_k has expected weight $\|v\|^2/D$, that is, $\mathbb{E}\mu_{A,v} = \|v\|^2 \mu_A$.
(d) The weights are v_k^2. For $v = e_1$, $\nu_{A,v} = \delta_{\lambda_1}$: this vector sees only λ_1. For $v = (1, \dots, 1)/\sqrt{D}$, all weights are $1/D$ and $\nu_{A,v} = \mu_A$.

Guided proof 4.14. (a) $T_m^0 e_1 = e_1$. If $T_m^j e_1$ is supported on the first $j+1$ positions, multiplying by a tridiagonal matrix extends the support by one.
(b) $j = 0$: $v_1 = V_m e_1$. If $A^j v_1 = V_m T_m^j e_1$ with $j \le m-2$, then $A^{j+1} v_1 = A V_m T_m^j e_1 = V_m T_m^{j+1} e_1 + \beta_m v_{m+1} (e_m^\top T_m^j e_1)$, and $e_m^\top T_m^j e_1 = 0$ because $T_m^j e_1$ is supported on the first $j+1 \le m-1$ positions. The identity thus holds for $j \le m-1$; at $j = m$ the vector $A^m v_1$ leaves $\mathcal{K}_m$ in general.
(c) For $i = 2j$, $j \le m-1$: $v_1^\top A^{2j} v_1 = \|A^j v_1\|^2 = \|V_m T_m^j e_1\|^2 = e_1^\top T_m^{2j} e_1$, since V_m has orthonormal columns and T_m is symmetric. For $i = 2j+1$, $j \le m-1$:

$$v_1^\top A^{2j+1} v_1 = (A^j v_1)^\top A (A^j v_1) = e_1^\top T_m^j V_m^\top A V_m T_m^j e_1 = e_1^\top T_m^{2j+1} e_1.$$

Together these cover $0 \le i \le 2m-1$.
(d) $e_1^\top T_m^i e_1 = e_1^\top S \Theta^i S^\top e_1 = \sum_j S_{1j}^2 \theta_j^i$. By Warm-up 4.11(a), $v_1^\top A^i v_1 = \int \lambda^i \, d\nu_{A,v_1}$. An m-point rule that is exact for all polynomials of degree $\le 2m-1$ is the Gauss rule.
(e) Lanczos produces $v_{j+1} = p_j(A) v_1$ with polynomials p_j of degree j, so $v_i^\top v_j = \int p_{i-1} p_{j-1} \, d\nu_{A,v_1}$. The diagonal entries

$$(T_m)_{jj} = v_j^\top A v_j = \int \lambda \, p_{j-1}(\lambda)^2 \, d\nu_{A,v_1}(\lambda)$$

involve degree at most $2m-1$, and the off-diagonal entries β_j, $j \le m-1$, are norms of $(A - \alpha_j) v_j - \beta_{j-1} v_{j-1}$, of degree at most $2m-2$. So T_m is a function of the moments $\int \lambda^i \, d\nu_{A,v_1}$, $i \le 2m-1$.

Guided proof 4.15. (a) $b = Ax$, so $y = q(A)Ax$ and $x - y = (I - Aq(A))x = p(A)x$. Conversely, if $p(0) = 1$, then $p(\lambda) - 1 = -\lambda q(\lambda)$ for a polynomial q of degree $< m$.
(b) $\|p(A)x\|_A^2 = x^\top p(A) A p(A) x = b^\top A^{-1} p(A) A p(A) A^{-1} b = b^\top p(A)^2 A^{-1} b$, because functions of A commute; this is $\int p^2 \lambda^{-1} \, d\mu_{A,b}$ by Warm-up 4.11(a).
(c) Fact 4.13 with (a) and (b). The right-hand side depends on A and b only through $\mu_{A,b}$.
(d) $p(\lambda) = \prod_{j=1}^r (1 - \lambda/\lambda_j)$ over the atoms has degree r, $p(0) = 1$, and vanishes on the support of $\mu_{A,b}$, so the error at step r is 0.

Exercise 4.16. $q(\eta f(A) g) = \frac{1}{2}\eta^2 \, g^\top f(A) A f(A) g - \eta \, g^\top f(A) g$. Write $a = g^\top f(A) g = \int f \, d\mu_{A,g}$ and $b = g^\top f(A) A f(A) g = \int \lambda f^2 \, d\mu_{A,g} > 0$. The minimum over η is at $\eta = a/b$, with value $-a^2/(2b)$, and $q(0) = 0$.

Guided proof 4.17. (a) Newton: $a = b = \int \lambda^{-1} d\mu_{A,g}$, decrease $\frac{1}{2} \int \lambda^{-1} d\mu_{A,g}$. Gradient: $a = \|g\|^2$, $b = \int \lambda \, d\mu_{A,g}$, decrease $\|g\|^4 / (2 \int \lambda \, d\mu_{A,g})$. The ratio is $\int \lambda^{-1} d\mu_{A,g} \int \lambda \, d\mu_{A,g} / \|g\|^4 = \mathbb{E}_\nu[\lambda^{-1}] \, \mathbb{E}_\nu[\lambda]$.
(b) $1 = (\mathbb{E}[\lambda^{1/2} \lambda^{-1/2}])^2 \le \mathbb{E}[\lambda] \, \mathbb{E}[\lambda^{-1}]$ by Cauchy–Schwarz.
(c) Both factors are ≥ 0 on $[m, M]$. Expanding, $-\lambda^2 + (m+M)\lambda - mM \ge 0$; divide by $\lambda > 0$.
(d) $\mathbb{E}\lambda + mM \, \mathbb{E}\lambda^{-1} \le m + M$, and $2\sqrt{mM \, \mathbb{E}\lambda \, \mathbb{E}\lambda^{-1}} \le \mathbb{E}\lambda + mM \, \mathbb{E}\lambda^{-1}$. So $\mathbb{E}\lambda \, \mathbb{E}\lambda^{-1} \le (m+M)^2/(4mM) = (\kappa+1)^2/(4\kappa)$. Equality holds for mass $\frac{1}{2}$ at each of m and M.

Exercise 4.18. $\mathbb{E}_\mu\lambda = cH_D/D$ and $\mathbb{E}_\mu\lambda^{-1} = \frac{1}{cD}\sum_k k = (D+1)/(2c)$, so the product is $H_D(D+1)/(2D) \approx \frac{1}{2}(\ln D + \gamma)$: 4.45 and 5.14. The spectrum would allow Newton to beat a tuned gradient step by a factor of about 1,025, but a gradient with uniform weights gains only 4.45. The weights, that is, how the gradient sits relative to the frame, decide.

Section 5

Warm-up 5.1. (a) $(a-\lambda)(b-\lambda)-\delta^2 = 0$ gives $\lambda = \frac{a+b}{2} \pm \sqrt{g^2/4+\delta^2}$. The top eigenvalue moves by $\sqrt{g^2/4+\delta^2} - g/2 \in [0,\delta]$, and the bottom one by the same amount. For $\delta \ll g$, $\sqrt{g^2/4+\delta^2} = \frac{g}{2}\sqrt{1+4\delta^2/g^2} \approx \frac{g}{2} + \delta^2/g$.
(b) The rows of the eigenvector equation are $a\cos\theta + \delta\sin\theta = \lambda_+\cos\theta$ and $\delta\cos\theta + b\sin\theta = \lambda_+\sin\theta$. Multiply the first by $\sin\theta$, the second by $\cos\theta$, and subtract: $(a-b)\sin\theta\cos\theta = \delta(\cos^2\theta - \sin^2\theta)$, that is, $\frac{g}{2}\sin 2\theta = \delta\cos 2\theta$.
(c) $g = \delta = 0.01$: the eigenvalues move by 0.0062 (0.6% of a), and $2\theta = \arctan 2$, so $\theta = 31.7°$.
(d) $\tan 2\theta \to \infty$, so $\theta \to 45°$ however small δ is. Once the gap is much smaller than the perturbation, the eigenvectors are those of E, not of A.

Guided proof 5.2. (a) For unit $v = \sum_{i\le k} z_i u_i$, $v^\top A v = \sum_{i\le k}\lambda_i z_i^2 \ge \lambda_k$, with equality at $v = u_k$.
(b) $\dim S + \dim \operatorname{span}(u_k,\dots,u_D) = k + (D-k+1) > D$.
(c) For unit $v = \sum_{i\ge k} z_i u_i$, $v^\top A v = \sum_{i\ge k}\lambda_i z_i^2 \le \lambda_k$. So every S has inner minimum at most λ_k, and the S of (a) attains it.

Guided proof 5.3. (a) $v^\top B v = v^\top A v + v^\top E v \le v^\top A v + \|E\|_{\mathrm{op}}$.
(b) For every S, $\min_S v^\top B v \le \min_S v^\top A v + \|E\|_{\mathrm{op}}$ (evaluate at the minimiser for A). Taking the maximum over S gives $\lambda_k(B) \le \lambda_k(A) + \|E\|_{\mathrm{op}}$. Exchanging A and B ($E \to -E$) gives the other inequality.

Exercise 5.5. If $i < j$ and $\pi(i) > \pi(j)$, then $b_{\pi(i)} \le b_{\pi(j)}$, and swapping the two values changes the sum by $a_i b_{\pi(j)} + a_j b_{\pi(i)} - a_i b_{\pi(i)} - a_j b_{\pi(j)} = (a_i - a_j)(b_{\pi(j)} - b_{\pi(i)}) \ge 0$. Each such swap removes at least one inversion, so finitely many swaps reach the identity without decreasing the sum.

Guided proof 5.6. (a) $\|A-B\|_{\mathrm{F}}^2 = \operatorname{tr}(A-B)^2 = \operatorname{tr}A^2 + \operatorname{tr}B^2 - 2\operatorname{tr}(AB)$, with $\operatorname{tr}A^2 = \sum_i a_i^2$ and $\operatorname{tr}B^2 = \sum_j b_j^2$.
(b) $\operatorname{tr}(AB) = \operatorname{tr}(U_A \operatorname{diag}(a) U_A^\top U_B \operatorname{diag}(b) U_B^\top) = \operatorname{tr}(\operatorname{diag}(a) W \operatorname{diag}(b) W^\top) = \sum_{i,j} a_i W_{ij} b_j W_{ij}$.
(c) The rows and columns of the orthogonal matrix W are unit vectors, so the rows and columns of P sum to 1.
(d) $\operatorname{tr}(AB) = \sum_{i,j} a_i b_j P_{ij}$ is linear in P, so it is at most its maximum over permutations, $\max_\pi \sum_i a_i b_{\pi(i)} = \sum_i a_i b_i$. Hence $\|A-B\|_{\mathrm{F}}^2 \ge \sum_i a_i^2 + \sum_i b_i^2 - 2\sum_i a_i b_i = \sum_i (a_i - b_i)^2$.

Exercise 5.7. Couplings of the two uniform measures are $\frac{1}{D}$ times doubly stochastic matrices, and the cost $\frac{1}{D}\sum_{i,j}(a_i - b_j)^2 P_{ij}$ is linear in P, so a permutation is optimal. For a permutation the cost is $\frac{1}{D}[\sum_i a_i^2 + \sum_i b_i^2 - 2\sum_i a_i b_{\pi(i)}]$, which the sorted coupling minimises. So $W_2(\mu_A,\mu_B)^2 = \frac{1}{D}\sum_i (a_i - b_i)^2 \le \|A-B\|_{\mathrm{F}}^2/D$ by Hoffman–Wielandt.

Guided proof 5.8. (a) $(A+E)v = \mu v$.
(b) $Auu^\top = \lambda_k uu^\top = uu^\top A$, so $AP = PA$.
(c) For $z = \sum_{j\ne k} z_j u_j$, $\|(A-\mu I)z\|^2 = \sum_{j\ne k}(\lambda_j - \mu)^2 z_j^2 \ge \min_{j\ne k}(\lambda_j - \mu)^2 \|z\|^2$. For $j \ne k$, $|\lambda_j - \mu| \ge |\lambda_j - \lambda_k| - |\lambda_k - \mu| \ge g_k - \|E\|_{\mathrm{op}}$, by Weyl.

(d) $(g_k - \|E\|_{\mathrm{op}})\|Pv\| \le \|(A-\mu I)Pv\| = \|P(A-\mu I)v\| = \|PEv\| \le \|E\|_{\mathrm{op}}$, and $\|Pv\|^2 = 1-(u^\top v)^2 = \sin^2\angle(u,v)$.
(e) If $\|E\|_{\mathrm{op}} \le g_k/2$, then $g_k - \|E\|_{\mathrm{op}} \ge g_k/2$ and (d) gives the claim. If $\|E\|_{\mathrm{op}} > g_k/2$, then $2\|E\|_{\mathrm{op}}/g_k > 1 \ge \sin\angle(u,v)$.

Exercise 5.10. (a) $h'(x) = \alpha(1+x)^{-\alpha-1} > 0$ and $h''(x) = -\alpha(\alpha+1)(1+x)^{-\alpha-2} < 0$. A concave function lies below its tangents: at x, $h(0) \le h(x) - xh'(x)$; at 0, $h(x) \le h(0) + xh'(0) = \alpha x$.
(b) $(\lambda_k - \lambda_{k+1})/\lambda_k = 1-(1+1/k)^{-\alpha} = h(1/k)$; insert $x = 1/k$.
(c) $\varphi(x) = cx^{-\alpha}$ is convex and decreasing, so $\varphi(k-1) - \varphi(k) \ge \varphi(k) - \varphi(k+1)$: the gap above λ_k is at least the gap below, and gaps to non-neighbours are larger still.

Guided proof 5.11. (a) Weyl: $|\Delta\lambda_k| \le \eta \le sck^{-\alpha}$ if and only if $k \le (sc/\eta)^{1/\alpha}$.
(b) By Exercise 5.10, $g_k = \lambda_k - \lambda_{k+1} \ge ck^{-\alpha}\frac{\alpha}{k}(1+\frac{1}{k})^{-\alpha-1} \ge \alpha c k^{-\alpha-1}2^{-\alpha-1}$.
(c) $\sin \le 2\eta/g_k \le 2^{\alpha+2}\eta k^{\alpha+1}/(\alpha c) \le s$ if and only if $k^{\alpha+1} \le s\alpha c/(2^{\alpha+2}\eta)$.
(d) $K_{\mathrm{vec}}/K_{\mathrm{val}} \propto \eta^{-1/(\alpha+1)}/\eta^{-1/\alpha} = \eta^{1/(\alpha(\alpha+1))} \to 0$.

Exercise 5.12. $K_{\mathrm{val}} = 0.5/\eta = 560$ and $K_{\mathrm{vec}} = (0.5/(8\eta))^{1/2} = 8.4$. The simulation resolves eigenvalues to rank 1,801 (about three times the guarantee) and eigenvectors to rank 169 (about twenty times the guarantee). A worst-case E puts its whole size $\|E\|_{\mathrm{op}} \approx 2\varepsilon\sqrt{D}$ on the pair of directions with the smallest gap. A random E spreads its size over all D^2 entries: each pair of eigenvectors feels only an entry $u_j^\top E u_k$ of size about ε, a factor of order $\sqrt{D}$ smaller.

Guided proof 5.13. (a) Product rule; the derivative of $A(\varepsilon)$ is E.
(b) The left side gives $u_k^\top E u_k + \lambda_k u_k^\top u_k'$ and the right side $\lambda_k' + \lambda_k u_k^\top u_k'$.
(c) $u_j^\top E u_k + \lambda_j u_j^\top u_k' = \lambda_k u_j^\top u_k'$, using $u_j^\top u_k = 0$.
(d) $2u_k^\top u_k' = 0$. Expand $u_k' = \sum_j (u_j^\top u_k') u_j$ and insert (c). Since $u_k(\varepsilon) = u_k + \varepsilon u_k' + O(\varepsilon^2)$ with $u_k' \perp u_k$, the component orthogonal to u_k has norm $\varepsilon\|u_k'\| + O(\varepsilon^2)$. In the eigenbasis, $u_j^\top E u_k = (\lambda_k - \lambda_j)\, u_j^\top u_k'$: the formula of Guided proof 1.81.8(d) with $W_{jk} = u_j^\top u_k'$.

Exercise 5.14. (a) With $Q = U$, $U^\top G U$ has the law of G: its off-diagonal entries $u_j^\top G u_k$ are independent $\mathcal{N}(0,1)$ and its diagonal entries $\mathcal{N}(0,2)$. By Guided proof 5.13, $\mathbb{E}\sin^2 \approx \varepsilon^2 \mathbb{E}\|\sum_{j\ne k} u_j^\top G u_k\, u_j/(\lambda_k - \lambda_j)\|^2 = \varepsilon^2 \sum_{j\ne k}(\lambda_k - \lambda_j)^{-2}$, and λ_k moves by $\varepsilon u_k^\top G u_k$, of standard deviation $\sqrt{2}\varepsilon$.
(b) For $|m| \ll k$, $\lambda_k - \lambda_{k+m} \approx -mg_k$, so these terms sum to $g_k^{-2}\sum_{m\ne 0} m^{-2} = 2\zeta(2)/g_k^2 = 2\zeta(2)k^{2\alpha+2}/(\alpha^2 c^2)$. For $j \gg k$, $\lambda_j \ll \lambda_k$ and each term is about $\lambda_k^{-2} = k^{2\alpha}/c^2$; there are about D of them.
(c) The ratio of the two parts is $2\zeta(2)k^2/(\alpha^2 D)$, which exceeds 1 for $k > \alpha\sqrt{D/(2\zeta(2))}$; for $\alpha = 1$ and $D = 2{,}000$ this is 25.
(d) $\varepsilon\sqrt{2\zeta(2)}\, k^{\alpha+1}/(\alpha c) = s$ gives the formula. With $\alpha = c = 1$, $s = \frac{1}{2}$, $\varepsilon = 10^{-5}$: $(0.5/(1.814 \times 10^{-5}))^{1/2} = 166$.

Compute 5.15. If $A = U\Lambda U^\top$, then $A + E = U(\Lambda + U^\top E U)U^\top$, and $U^\top E U$ has the law of E: the eigenvalues have the law of those of $\Lambda + E$, and the eigenvectors are rotated by U, which leaves all angles unchanged.

```
rng = np.random.default_rng(0); D = 2000; eps = 1e-5
lam = 1/np.arange(1, D+1)
z = rng.standard_normal((D, D)); E = eps*(z+z.T)/np.sqrt(2)
w, v = np.linalg.eigh(np.diag(lam)+E); w, v = w[::-1], v[:, ::-1]
```

```
rel = np.abs(w-lam)/lam                              # (a)
sines = np.sqrt(np.clip(1-np.diag(v)**2, 0, None))   # (b)
```

Guided proof 5.16. (a) $U_A^\top(A-B)U_B = \operatorname{diag}(a)U_A^\top U_B - U_A^\top U_B \operatorname{diag}(b) = \operatorname{diag}(a)W - W\operatorname{diag}(b)$, with entries $(a_i-b_j)W_{ij}$.
(b) The same computation with $f(A) = U_A \operatorname{diag}(f(a))U_A^\top$ and $f(B) = U_B \operatorname{diag}(f(b))U_B^\top$.
(c) $\|f(A)-f(B)\|_{\mathrm{F}}^2 = \sum_{i,j}(f(a_i)-f(b_j))^2 W_{ij}^2 \le L^2 \sum_{i,j}(a_i-b_j)^2 W_{ij}^2 = L^2\|A-B\|_{\mathrm{F}}^2$.

Exercise 5.17. $|f'(\lambda)| = \nu(\lambda+\delta)^{-\nu-1} \le \nu\delta^{-\nu-1}$ on $[0,\infty)$; the mean value theorem gives the Lipschitz constant, and Guided proof 5.16 the matrix bound.

Exercise 5.18. With $P = VV^\top$ and $P' = V'V'^\top$: $\|P-P'\|_{\mathrm{F}}^2 = \operatorname{tr}P + \operatorname{tr}P' - 2\operatorname{tr}(PP') = 2k - 2\operatorname{tr}(PP')$, and $\operatorname{tr}(PP') = \operatorname{tr}(V'^\top VV^\top V') = \|V^\top V'\|_{\mathrm{F}}^2 = \sum_i \cos^2\theta_i$, which also equals $\|PP'\|_{\mathrm{F}}^2$. So $\|P-P'\|_{\mathrm{F}}^2 = 2\sum_i(1-\cos^2\theta_i) = 2\sum_i \sin^2\theta_i$. $\Pi_k(A) = f(A)$ with f the indicator of $[\lambda_k,\infty)$, which jumps; its stability comes from the gap at the cut, through Remark 5.9.

Warm-up 5.20. $\|A - UMU^\top\|_{\mathrm{F}} = \|U^\top AU - M\|_{\mathrm{F}}$. A diagonal M changes only diagonal entries; the best choice is $M = \operatorname{Diag}(U^\top AU)$, which leaves the off-diagonal entries $u_i^\top Au_j$. Also $\operatorname{tr}\mathcal{P}_U(A) = \sum_i u_i^\top Au_i = \operatorname{tr}(U^\top AU) = \operatorname{tr}A$.

Guided proof 5.21. (a) $x_i^\top Ax_i = \sum_j \lambda_j (u_j^\top x_i)^2$; sum over i and use $\sum_i (u_j^\top x_i)^2 = \|X^\top u_j\|^2$.
(b) $X^\top u_j$ holds the coefficients of the projection of the unit vector u_j onto $\operatorname{span}(x_1,\dots,x_k)$, so $c_j \le 1$ (Bessel). $\sum_j c_j = \sum_i \sum_j (u_j^\top x_i)^2 = \sum_i \|x_i\|^2 = k$.
(c) $\sum_j \lambda_j c_j - \sum_{j\le k}\lambda_j = \sum_{j\le k}\lambda_j(c_j-1) + \sum_{j>k}\lambda_j c_j \le \lambda_k \sum_{j\le k}(c_j-1) + \lambda_k \sum_{j>k} c_j = \lambda_k(\sum_j c_j - k) = 0$, using $c_j - 1 \le 0$ and $\lambda_j \ge \lambda_k$ for $j \le k$, and $\lambda_j \le \lambda_k$ for $j > k$.
(d) Apply (c) to the k columns of U with the largest Rayleigh quotients. For $k = D$ both sums equal $\operatorname{tr}A$.
(e) $k = 1$ gives the maximum. For the minimum, subtract the case $k = D-1$ from the equal totals.

Section 6

Guided proof 6.1. (a) $U'^\top U + U^\top U' = 0$.
(b) $A' = U'\Lambda U^\top + U\Lambda' U^\top + U\Lambda U'^\top$, so $U^\top A'U = W\Lambda + \Lambda' + \Lambda W^\top = \Lambda' + W\Lambda - \Lambda W$.
(c) The diagonal of $W\Lambda - \Lambda W$ is zero, so $\lambda_k' = (U^\top A'U)_{kk} = u_k^\top A' u_k$. Off the diagonal, $(U^\top A'U)_{ij} = (\lambda_j - \lambda_i)W_{ij}$. With $E = A'$ and $W_{jk} = u_j^\top u_k'$, this is Guided proof 5.13(b) and (c).
(d) If the spectrum is stable, $\Lambda' \approx 0$ and $U^\top A'U$ is nearly off-diagonal; its entries divided by the gaps are the rotation rates.

Exercise 6.2. (a) $hW_{ij} \approx u_i^\top \Delta A u_j/(\lambda_j - \lambda_i) = \sigma\sqrt{h}\, u_i^\top G u_j/(\lambda_j-\lambda_i)$, and $u_i^\top G u_j \sim \mathcal{N}(0,1)$ by Exercise 5.14(a): variance $\sigma^2 h/(\lambda_i-\lambda_j)^2$. The eigenvalue changes by $\sigma\sqrt{h}\, u_k^\top G u_k$, with variance $2\sigma^2 h$.
(b) $(U^\top A'U)_{ij} = (\lambda_j - \lambda_i)W_{ij}$ now grows with the gap. Nearly tied pairs rotate as fast as all others but barely change the matrix; well separated pairs produce the largest entries.
(c) Measure the rotation of eigenvector pairs, or the loss of capture of single eigenvectors, over a fixed lag and plot it against the gap. Isotropic noise gives slope -2 in log–log coordinates (for the squared rotation); frame diffusion gives a flat line. Replays that save eigenvectors at nearby steps give this directly.

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